

SCE 2026 · GROUP 04 · TMP: PITCH

Walking with meaning

The 16 current-concept chapters of the Foundation and Specification, in full, as they stand in tmp-pitch. Index pages and the superseded Morning Care pages are left out.

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a1.1 Garden walks, alone and in pairs

Abstract. Two activities, both walks: alone (SA-01) and with a fellow resident (SA-02). Fresh air and movement stimulate memory and steady emotions; company gives a reason to go. Breakdowns on walks alone: nobody to talk to, staff unavailable, confusion and attempts to go home. On pair walks: a slip the partner does not notice, a partner who does not fit, overexertion while talking. Stumbles while talking can happen on both.

Situated Activity - A walk on your own [SA-01]

Report Format: §1.1 Operational Demands

Fields

Field	Description
ID	SA-01
Name	A walk on your own
Status	In Progress
Evidence basis	Provisional. Literature and group reasoning only. - Forbes et al. (2015), Cochrane review: exercise may improve daily functioning in dementia - Bratman et al. (2015): a walk in nature lowered rumination (healthy adults) - Detweiler et al. (2008): an enclosed garden on a dementia unit; fewer agitated behaviours (one unit, before-after) - Victor (2012): loneliness in care homes is common - Lai & Arthur (2003): wandering and exit-seeking in people with dementia
Goal	Go out when you like: fresh air, movement, a change of scene. Back safely.
Performing Actor(s)	- ST-01: Resident - walks alone - ST-04: Care worker - opens the door; notices who is out - ST-06: Physiotherapist - assesses walking unaccompanied
Trigger	Restless; fine weather; garden door open; an old habit
Constituent Tasks	1. Decide to go 2. Find the door; coat, shoes, rollator 3. Walk the loop; look, listen 4. Rest on a bench 5. Notice when to turn back: tired, cold 6. Find the door; let staff know
Context	Only residents assessed fit (ST-06). Enclosed garden loop, one coded gate. Nobody to talk to; staff glance out of the window. The walk depends on staff: to open the door, keep an eye out, or walk along when walking alone is not allowed. Confused, some residents set out for a former home.
Frequency and duration	Varies widely; 5-30 min; mostly in fine weather
Tools and artifacts	Garden door, coded gate, loop path, benches, rollator, a window onto the garden

Existing workarounds	• Loop that leads back to the door • Door open in fine weather; staff keep an eye out • Accompanied walk when walking alone is not allowed • Coded gate; staff redirect a resident heading for the exit
Observed breakdowns	• Nobody to talk to: lonely; stops going out (PS-06) • Staff unavailable: the door stays closed, nobody keeps an eye out, the walk does not happen • Confused: forgets where or why; heads for the gate to go home • Fall or stumble, unseen; tired, far from the door

Graph edges

performed-by → ST-01 (supported by ST-04, ST-06); extended-by → SA-02; illustrated-by → PS-06; setting of → PS-02 (a consideration, not a breakdown).

Situated Activity - A walk with a fellow resident [SA-02]

Report Format: §1.1 Operational Demands

Fields

Field	Description
ID	SA-02
Name	A walk with a fellow resident
Status	In Progress
Evidence basis	Provisional. Literature and group reasoning only. • de Medeiros et al. (2012): residents with dementia form and keep friendships • Orange et al. (1996): in Alzheimer's disease, trouble in conversation grows and repair succeeds less often • Schegloff et al. (1977): speakers prefer to fix their own slips • Bora et al. (2015), meta-analysis: theory of mind is impaired in Alzheimer's disease • Clare (2003): people aware of decline protect themselves, or face it • Lundin-Olsson et al. (1997): older people who stop walking when they talk fall more often • Montero-Odasso et al. (2012): with cognitive impairment, walking slows and grows unsteady under a second task
Goal	Company as the reason to go out; a conversation both enjoy
Performing Actor(s)	• ST-01: Resident - walks, talks • ST-02: Fellow resident - walks, talks • ST-05: Activity coordinator - matches pairs • ST-04: Care worker - starts the walk; keeps an eye out
Trigger	Staff see two residents who get along; one invites the other; a pair's habit

Constituent Tasks	1. Match: pace, mood, stage of dementia, who gets along with whom 2. Ask both 3. Introduce, or re-introduce 4. Walk side by side, at one pace 5. Talk: what they see, what they share 6. After a slip: give time; then the last topic, a new one, or the surroundings 7. Rest on a bench 8. One wants to go back: both turn
Context	The match makes or breaks the walk. Staff know some shared pasts, and who dislikes whom, informally. Names are forgotten. Slips grow more frequent as dementia progresses: a lost thread, a missing word, a story told twice, a belief from another time ("My mother is waiting"). Speakers prefer to repair their own slips; a partner with dementia often misses them. Many residents know they slipped, and feel it. Talking slows walking, hides tiredness and raises fall risk.
Frequency and duration	Occasional; 15-30 min
Tools and artifacts	Staff knowledge, care record, life-story book, garden, benches
Existing workarounds	• Pair residents who sit together at meals, or share a past • Re-introduce them at each walk • Same pair, same time, same route • Care worker walks along • Gentle redirection by staff: to the present, to the feeling, to a new topic
Observed breakdowns	• No partner; mismatch in pace or mood • Two residents who dislike each other, paired by someone who did not know (PS-03) • Agrees to please staff or the partner: "I suppose so" (PS-03) • Loses the thread; the partner does not notice; the talk dies (PS-01) • Helped too fast, corrected or quizzed ("You told me already", "Do you remember...?"): embarrassed • After repeated slips: avoids the next walk • A personal fact, used by staff to start a conversation, passed on (PS-05) • Overexertion: distracted by the partner, walks past tiredness and the set time (PS-04) • Stumbles while talking (PS-01, PS-04)

Graph edges

performed-by → ST-01, ST-02 (supported by ST-04, ST-05); extends → SA-01; illustrated-by → PS-01, PS-03 to PS-05.

The two walks compared

Aspect	SA-01 Alone	SA-02 With a fellow resident
Company	None	Present; often has dementia too
Notices a slip	Nobody	The partner, often not
Notices a stumble	Whoever looks out	The partner, if able
Notices tiredness	The resident, if at all	Nobody: the talk is good

References

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a2.1 Residents, family and staff

Abstract. Seven stakeholders. Two walk and talk: the resident (ST-01) and a fellow resident (ST-02). Three support the walk: the relative, who tells the life story and agrees what may be said (ST-03); the care worker, who answers alerts and painful questions (ST-04); the activity coordinator, who keeps pairs, settings and the relationship map (ST-05). Two set limits: medical and therapy staff (ST-06) and management (ST-07). Shared: the resident's well-being and trust. Divided on: freedom against safety, and how much the System may know and pass on.

Stakeholder - Resident with dementia [ST-01]

Report Format: §1.1 Operational Demands

Fields

Field	Description
ID	ST-01
Name	Resident with dementia
Status	In Progress
Evidence basis	Provisional. Group reasoning only. Derived from SA-01, SA-02.
Type	Direct - walks, talks, consents, sets own preferences
Values and motivations	- HV-06 Social connectedness (high) - company, a chat - HV-02 Dignity (high) - slips not remarked on; never walked like a pet - HV-08 Truthfulness (high) - the truth, from someone trusted - HV-01 Autonomy (high) - own reasons, own settings, own consent - HV-03 Privacy (high) - own conversations and secrets - HV-07 Identity (high) - own past, own words - HV-05 Well-being (high) - fresh air, calm - HV-04 Safety (medium) - no falls, no getting lost
Relationship to other Stakeholders	Walks alone, or with ST-02. Family (ST-03) tells the life story. Helped by ST-04. Agrees settings with ST-05. Walking assessed by ST-06.

Graph edges

performs → SA-01, SA-02; holds → HV-01 to HV-08; appears-in → PS-01 to PS-06; elaborated-as → HP-01, HP-02 (§2.2).

Stakeholder - Fellow resident [ST-02]

Report Format: §1.1 Operational Demands

Fields

Field	Description
ID	ST-02
Name	Fellow resident (walking partner)
Status	In Progress

Evidence basis	Provisional. Group reasoning only. Derived from SA-02.
Type	Direct - walks and talks with ST-01; often has dementia too
Values and motivations	- HV-06 Social connectedness (high) - a mate to talk to - HV-01 Autonomy (high) - free to decline or end a walk - HV-03 Privacy (high) - own secrets stay with own System - HV-05 Well-being (high) - own walk, own health - HV-02 Dignity (medium) - a partner, not a minder
Relationship to other Stakeholders	Matched with ST-01 by ST-05; both consent. Own relatives, own care plan, own copy of the System.

Graph edges

performs → SA-02; holds → HV-01 to HV-03, HV-05, HV-06; appears-in → PS-01, PS-03 to PS-05; elaborated-as → HP-01, HP-02 (§2.2).

Stakeholder - Relative [ST-03]

Report Format: §1.1 Operational Demands

Fields

Field	Description
ID	ST-03
Name	Relative
Status	In Progress
Evidence basis	Provisional. Group reasoning only. Derived from SA-01, SA-02.
Type	Indirect - tells the life story; agrees sensitive topics, answers and settings; may consent on the resident's behalf
Values and motivations	- HV-06 Social connectedness (high) - their relative has company - HV-08 Truthfulness (high) - no lies told to their relative - HV-05 Well-being (high) - their relative calm - HV-07 Identity (high) - their relative stays the person they knew - HV-03 Privacy (low) - may want to know more than the resident would share
Relationship to other Stakeholders	Relative of ST-01. Gives life story and sensitive topics to ST-05. Agrees with ST-04 who answers what. May consent on ST-01's behalf.

Graph edges

supports → SA-01, SA-02; holds → HV-03, HV-05 to HV-08; appears-in → PS-02, PS-05.

Stakeholder - Care worker [ST-04]

Report Format: §1.1 Operational Demands

Fields

Field	Description
ID	ST-04

Name	Care worker
Status	In Progress
Evidence basis	Provisional. Group reasoning only. Derived from SA-01, SA-02.
Type	Direct - starts walks, answers alerts; trusted person for named sensitive topics
Values and motivations	- HV-04 Safety (high) - every resident back, unharmed - HV-08 Truthfulness (high) - painful news told as agreed with the family - HV-05 Well-being (high) - residents out, and calm - HV-03 Privacy (medium) - for residents; no surveillance of own work
Relationship to other Stakeholders	Trusted person for named topics of named residents. Keeps the relationship map current with ST-05. Follows ST-06 on walking and medical news.

Graph edges

supports → SA-01, SA-02; holds → HV-03 to HV-05, HV-08; appears-in → PS-01 to PS-06; elaborated-as → HP-03 (§2.2).

Stakeholder - Activity coordinator [ST-05]

Report Format: §1.1 Operational Demands

Fields

Field	Description
ID	ST-05
Name	Activity coordinator
Status	In Progress
Evidence basis	Provisional. Group reasoning only. Derived from SA-02.
Type	Direct - keeps profiles, pairs, per-person settings and the relationship map
Values and motivations	- HV-06 Social connectedness (high) - pairs that work - HV-07 Identity (high) - profiles that fit the person - HV-03 Privacy (high) - map and profiles in their care - HV-01 Autonomy (medium) - settings agreed with the resident
Relationship to other Stakeholders	Writes profiles with ST-01 and ST-03. Pairs ST-01 and ST-02. Reviews the map with ST-04 after each event. Employed by ST-07.

Graph edges

supports → SA-01, SA-02; holds → HV-01, HV-03, HV-06, HV-07; appears-in → PS-03, PS-06.

Stakeholder - Medical and therapy staff [ST-06]

Report Format: §1.1 Operational Demands

Fields

Field	Description
ID	ST-06

Name	Medical and therapy staff: physician, nurse, physiotherapist, psychologist
Status	In Progress
Evidence basis	Provisional. Group reasoning only. Derived from SA-01, SA-02.
Type	Indirect - assess fall risk and walking alone; set walk length; tell medical news
Values and motivations	- HV-04 Safety (high) - fall risk - HV-05 Well-being (high) - HV-08 Truthfulness (high) - a diagnosis is told by them - HV-03 Privacy (medium) - medical confidentiality
Relationship to other Stakeholders	Assess ST-01 and ST-02. Advise ST-04 and ST-05. Trusted person for medical topics.

Graph edges

supports → SA-01, SA-02; holds → HV-03 to HV-05, HV-08; appears-in → PS-04.

Stakeholder - Care home management [ST-07]

Report Format: §1.1 Operational Demands

Fields

Field	Description
ID	ST-07
Name	Care home management
Status	In Progress
Evidence basis	Provisional. Group reasoning only. Derived from SA-01.
Type	Indirect - data controller; exits; local network; decides on adoption
Values and motivations	- HV-04 Safety (high) - liable for falls and exits - HV-03 Privacy (high) - GDPR, health data - HV-05 Well-being (medium) - quality of care
Relationship to other Stakeholders	Employs ST-04 and ST-05. Responsible for the garden, the exits and the local server.

Graph edges

governs → SA-01 (exits), System data (local only); holds → HV-03 to HV-05.

a3.1 What goes wrong on a walk

Abstract. Six scenarios. On pair walks: a slip nobody noticed (PS-01), a pairing nobody wanted (PS-03), talk that hides tiredness (PS-04), a secret used to start a conversation (PS-05). On walks alone: nobody to talk to (PS-06). On any walk, a consideration rather than a breakdown: a painful question, answered by the wrong person (PS-02). In each, something the walk depends on (memory, consent, attention, company, trust) is left to chance.

Problem Scenario - A slip nobody noticed [PS-01]

Report Format: §1.1 Operational Demands

Fields

Field	Description
ID	PS-01
Name	A slip nobody noticed
Status	In Progress
Evidence basis	Provisional. Constructed from SA-02 and the a2.1 records. No resident or care worker has confirmed it. • Orange et al. (1996): repair succeeds less often in Alzheimer's disease • Bora et al. (2015): theory of mind impaired in Alzheimer's disease • Lundin-Olsson et al. (1997): stopping to talk predicts falls
Narrative	After lunch two residents walk the garden loop; a care worker paired them. She ran a post office counter and is telling him about pension day. Mid-sentence the word goes, then the story. He has dementia too, does not notice, and talks about the weather. She falls silent. Pointing at the pond, he catches his foot on the edge of the path; she grabs his arm. At the bench she says: "My head's no good any more," and asks to go in. The next day she is "too tired" to walk.
Stakeholders involved	• ST-01: Resident with dementia • ST-02: Fellow resident • ST-04: Care worker
Activities involved	SA-02: A walk with a fellow resident
Breakdown point	A slip that nobody repairs, and that costs face
Causes	Root causes: • Memory and word-finding loss • The partner's weakened theory of mind: the slip goes unnoticed • Nobody keeps the thread Contributing causes: • Walking and talking at once • Fatigue • Awareness of decline

Sub-problems	1. Detection: nobody notices the slip 2. Recovery: no way back into the story 3. Face: the slip exposes her 4. Safety: talking on the move, a stumble 5. Withdrawal: she avoids the next walk
Out of scope	• Speech and language therapy - ST-06 • Pairing without staff and without both residents' consent • Treating low mood - ST-06

Graph edges

illustrates → Problem; involves → ST-01, ST-02, ST-04, SA-02; reimagined-as → DS-01 (§2.1).

Problem Scenario - The wrong answer [PS-02]

Report Format: §1.1 Operational Demands

Fields

Field	Description
ID	PS-02
Name	The wrong answer
Status	In Progress
Evidence basis	Provisional. Set on a walk alone (SA-01): a consideration for the design, not a breakdown of the walk. No resident or care worker has confirmed it. • Mental Health Foundation (2016): truth and lies in dementia care
Narrative	A resident walks the garden alone. Her husband died last year. A new care worker passes, and she asks: "Is Thomas coming today?" Nobody told the worker, who says: "Your husband passed away, didn't you know?" She weeps as if hearing it for the first time. Last week another worker said: "He's at work." She now trusts neither.
Stakeholders involved	• ST-01: Resident with dementia • ST-04: Care worker • ST-03: Relative (agreement not passed on)
Activities involved	SA-01: A walk on your own
Breakdown point	A painful question answered by someone uninformed; earlier, an untrue answer
Causes	Root causes: • The death is not retained • The question goes to whoever is near • Agreements kept in staff heads Contributing causes: • Staff turnover • Loneliness on walks alone

Sub-problems	1. Detection: the question is sensitive 2. Routing: who should answer 3. Truth: a truthful answer, from a trusted person 4. Trust: a kind lie, once found out, costs it
Out of scope	• Grief counselling - ST-06 • Deciding what is told - care team with family

Graph edges

illustrates → Problem; involves → ST-01, ST-03, ST-04, SA-01; reimagined-as → DS-02 (§2.1).

Problem Scenario - A pairing nobody wanted [PS-03]

Report Format: §1.1 Operational Demands

Fields

Field	Description
ID	PS-03
Name	A pairing nobody wanted
Status	In Progress
Evidence basis	Provisional. Constructed from SA-02 and the a2.1 records. Group reasoning only; no resident or care worker has confirmed it.
Narrative	A new care worker sees two residents sitting alone and suggests, in front of both, that they walk together. He says "I suppose so"; she does not answer, and goes along. Last month they argued over a chair in the lounge. The regular staff know; the new worker does not. On the loop he corrects her twice. She walks faster, then sits on a bench and refuses to go on. The care worker is called from the ward to bring them in, one at a time.
Stakeholders involved	• ST-01: Resident with dementia • ST-02: Fellow resident • ST-04: Care worker • ST-05: Activity coordinator (pairs not recorded)
Activities involved	SA-02: A walk with a fellow resident
Breakdown point	A pairing made by someone who did not know their history, agreed to under pressure
Causes	Root causes: • Who dislikes whom is known informally, not recorded • Asked together, "no" is hard to say • The feeling of a quarrel outlasts the memory of its cause Contributing causes: • Staff turnover • The wish to get residents out • No easy way to end a pair walk
Sub-problems	1. Knowledge: the history between two residents is not recorded 2. Consent: asked together, one goes along to please 3. Conflict: correction and irritation on the walk 4. Exit: no graceful way to end the walk

Out of scope	- Resolving conflicts between residents - ST-04, ST-05 - Where residents live and sit
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Graph edges

illustrates → Problem; involves → ST-01, ST-02, ST-04, ST-05, SA-02; addressed-by → F1, IDP-02, TECH-05.

Problem Scenario - One more story [PS-04]

Report Format: §1.1 Operational Demands

Fields

Field	Description
ID	PS-04
Name	One more story
Status	In Progress
Evidence basis	Provisional. Constructed from SA-02 and the a2.1 records. No resident or physiotherapist has confirmed it. - Lundin-Olsson et al. (1997): stopping to talk predicts falls - Montero-Odasso et al. (2012): walking slows and grows unsteady under a second task
Narrative	Two residents are on their third loop, deep in stories about the town's old dance hall. His physiotherapist set twenty minutes; they are at forty. He slows and leans on his rollator, but keeps talking. Turning to her mid-sentence, he misses the edge of the path and goes down on one knee. Nobody is near. Back inside he is exhausted and sleeps through dinner.
Stakeholders involved	- ST-01: Resident with dementia - ST-02: Fellow resident - ST-04: Care worker - ST-06: Physiotherapist
Activities involved	SA-02: A walk with a fellow resident
Breakdown point	Good talk hides tiredness and draws attention from the path
Causes	Root causes: - Walking and talking share limited attention - Engaged talk masks fatigue - Nobody on the walk keeps time or watches pace Contributing causes: - The walk length is in the care plan, not on the walk - Turning to face the partner while walking - A rollator in one hand, gestures with the other
Sub-problems	- Attention: talk takes it from the path - Fatigue: not noticed in time - Time: the walk overruns - Help: a fall, out of sight of staff

Out of scope	- Fall-risk assessment and training - ST-06 - Changes to paths and benches - ST-07
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Graph edges

illustrates → Problem; involves → ST-01, ST-02, ST-04, ST-06, SA-02; addressed-by → F3, F5; reimagined-as → DS-01 (in part: the bench, the head-back cue).

Problem Scenario - A secret passed on [PS-05]

Report Format: §1.1 Operational Demands

Fields

Field	Description
ID	PS-05
Name	A secret passed on
Status	In Progress
Evidence basis	Provisional. Constructed from SA-02 and the a2.1 records. Group reasoning only; no resident or care worker has confirmed it.
Narrative	To get two residents talking, a care worker introduces them: "You have something in common: you both lost your husbands this year." One of them had told only her daughter how hard that was, and wanted it kept that way. On the next walk, with a third resident nearby, the other asks her whether she still cries at night. She goes inside, and stops walking with anyone.
Stakeholders involved	- ST-01: Resident with dementia - ST-02: Fellow resident - ST-04: Care worker - ST-03: Relative (told in confidence)
Activities involved	SA-02: A walk with a fellow resident
Breakdown point	A personal fact used to start a conversation, without its owner's consent
Causes	Root causes: - Staff use what they know to connect residents - No rule on what may be shared, and with whom - A partner with dementia repeats what she hears, without a sense of what is private Contributing causes: - The wish to find common ground quickly - Weakened theory of mind: the partner does not see the hurt
Sub-problems	- Consent: nobody asked the owner - Scope: told to one person, repeated to others - Trust: she withdraws from company
Out of scope	- What residents choose to tell each other themselves - Confidentiality of care records - ST-07

Graph edges

illustrates → Problem; involves → ST-01, ST-02, ST-03, ST-04, SA-02; addressed-by → F1, IDP-02, TECH-05.

Problem Scenario - Nobody to talk to [PS-06]

Report Format: §1.1 Operational Demands

Fields

Field	Description
ID	PS-06
Name	Nobody to talk to
Status	In Progress
Evidence basis	Provisional. Constructed from SA-01 and the a2.1 records. No resident or care worker has confirmed it. • Victor (2012): loneliness in care homes is common
Narrative	For years he walked his dog twice a day and chatted at every gate. In the care home he walks the garden loop alone. Nobody to talk to; he tells the bench the same story twice. The walks get shorter. By winter he no longer goes out, and sits by the window. Staff notice only when his mood drops and his legs weaken.
Stakeholders involved	• ST-01: Resident with dementia • ST-04: Care worker • ST-05: Activity coordinator
Activities involved	SA-01: A walk on your own
Breakdown point	Without company the walk loses its reason, and the resident withdraws
Causes	Root causes: • Loneliness • No partner who fits; family far away • The walk had a purpose (the dog, the neighbours) that is gone Contributing causes: • Staff time • Gradual withdrawal is hard to see • Fear of slips in company
Sub-problems	1. Company: nobody to talk to 2. Reason: the walk loses its purpose 3. Notice: withdrawal is seen late 4. Bridge: no route from walking alone to walking with someone
Out of scope	• Treating low mood - ST-06 • Visiting arrangements of families

Graph edges

illustrates → Problem; involves → ST-01, ST-04, ST-05, SA-01; addressed-by → F1 (bridge), F2 (conversation partner, where agreed); reimagined-as → DS-02 (in part: the guide as company).

References

- Bora et al. (2015); Lundin-Olsson et al. (1997); Mental Health Foundation (2016); Montero-Odasso et al. (2012); Orange et al. (1996); Victor (2012). See a1.1 References.

b1.1 What residents and staff value

Abstract. Eight values. Social connectedness leads: company is the reason to walk and to talk. Dignity includes face: a slip is not exposed. Truthfulness: the System does not deceive, corrects its own errors, and leaves painful truths to a trusted person. Privacy covers other people's secrets as well as one's own.

Human Value - Autonomy [HV-01]

Report Format: §1.2 Human Values and Human Factors

Fields

Field	Description
ID	HV-01
Name	Autonomy
Status	In Progress
Evidence basis	Provisional. Literature and group reasoning only.
Situated meaning	Walk or not, with whom, how long. Choose own settings. Consent before joining a person and before each conversation starter.
Held by	ST-01, ST-02 (high); ST-05 (medium)
Intrinsic or instrumental	Intrinsic. Also supports Dignity.
Recognition criteria	Served: the resident sets and changes their own settings; a "no" ends the offer, without a reason asked. Violated: walks arranged for staff convenience; a starter used without asking; persuasion; reminders of missed walks.
Source	Ryan, R. M. & Deci, E. L. (2000). Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. <i>American Psychologist</i> , 55(1), 68-78.

Human Value - Dignity [HV-02]

Report Format: §1.2 Human Values and Human Factors

Fields

Field	Description
ID	HV-02
Name	Dignity
Status	In Progress
Evidence basis	Provisional. Literature and group reasoning only.
Situated meaning	Spoken to as an adult. Slips are not remarked on. No quizzing, no correcting, no praise for walking. A walk taken for the resident's sake, never as a routine run.
Held by	ST-01 (high); ST-02 (medium)
Intrinsic or instrumental	Intrinsic.

Recognition criteria	Served: help after a slip offered to both walkers, as a statement; the slip not mentioned afterwards. Violated: "You told me that already"; "Do you remember...?"; finishing someone's sentence; being "taken out" like a dog.
Source	Kitwood, T. (1997). <i>Dementia Reconsidered: The Person Comes First</i> . Open University Press. Goffman, E. (1967). <i>Interaction Ritual</i> . Anchor Books.

Human Value - Privacy [HV-03]

Report Format: §1.2 Human Values and Human Factors

Fields

Field	Description
ID	HV-03
Name	Privacy
Status	In Progress
Evidence basis	Provisional. Literature and group reasoning only.
Situated meaning	Own conversations, own secrets, own whereabouts. Another resident's secrets never reach you through the System. Data stays in the care home. Staff: no surveillance of their own work.
Held by	ST-01, ST-02, ST-05, ST-07 (high); ST-04, ST-06 (medium); ST-03 (low - may want more than the resident would share)
Intrinsic or instrumental	Intrinsic. Also supports Dignity.
Recognition criteria	Served: nothing leaves the home; only consented items exchanged; no audio stored. Violated: a secret said to the wrong person; conversation recorded; route logged.
Source	Friedman, B., Kahn, P. H., Jr. & Borning, A. (2008). Value Sensitive Design and Information Systems. In K. E. Himma & H. T. Tavani (Eds.), <i>The Handbook of Information and Computer Ethics</i> . Wiley. Niemeijer et al. (2010).

Human Value - Safety [HV-04]

Report Format: §1.2 Human Values and Human Factors

Fields

Field	Description
ID	HV-04
Name	Safety
Status	In Progress
Evidence basis	Provisional. Literature and group reasoning only.
Situated meaning	No falls, no getting lost; help arrives quickly. Talking while walking raises the risk.
Held by	ST-04, ST-06, ST-07 (high); ST-01 (medium)
Intrinsic or instrumental	Instrumental: the condition for walking at all.

Recognition criteria	Served: stumbles and tiredness noticed; alerts reach a person. Violated: the walk continues past tiredness; a fall goes unanswered.
Source	Härlein, J. et al. (2009). Fall risk factors in older people with dementia or cognitive impairment. <i>Journal of Advanced Nursing</i> , 65(5), 922-933. Lundin-Olsson et al. (1997).

Human Value - Well-being [HV-05]

Report Format: §1.2 Human Values and Human Factors

Fields

Field	Description
ID	HV-05
Name	Well-being
Status	In Progress
Evidence basis	Provisional. Literature and group reasoning only.
Situated meaning	Fresh air, movement, calm. Emotional safety: free from embarrassment, conflict, pressure, and painful news at the wrong moment.
Held by	ST-01 to ST-04, ST-06 (high); ST-07 (medium)
Intrinsic or instrumental	Intrinsic.
Recognition criteria	Served: pleasure on the walk; calm afterwards. Violated: distress after a slip, a clash or a blunt answer.
Source	Forbes, D. et al. (2015). Exercise programs for people with dementia. <i>Cochrane Database of Systematic Reviews</i> , (4), CD006489. Bratman et al. (2015).

Human Value - Social connectedness [HV-06]

Report Format: §1.2 Human Values and Human Factors

Fields

Field	Description
ID	HV-06
Name	Social connectedness
Status	In Progress
Evidence basis	Provisional. Literature and group reasoning only.
Situated meaning	Company and conversation: a fellow resident, or the guide where agreed. Less lonely.
Held by	ST-01, ST-02, ST-03, ST-05 (high)
Intrinsic or instrumental	Both. Valued in itself; also the reason to go out.
Recognition criteria	Served: more walks with company; conversations resume after slips. Violated: a device in place of people; partners withdraw after slips.
Source	Victor, C. R. (2012). Loneliness in care homes: a neglected area of research? <i>Aging Health</i> , 8(6), 637-646. Holwerda et al. (2014).

Human Value - Identity [HV-07]

Report Format: §1.2 Human Values and Human Factors

Fields

Field	Description
ID	HV-07
Name	Identity
Status	In Progress
Evidence basis	Provisional. Literature and group reasoning only.
Situated meaning	Remain who you were: trade, places, people, own words and dialect.
Held by	ST-01, ST-03, ST-05 (high)
Intrinsic or instrumental	Intrinsic. Also gives conversation a subject.
Recognition criteria	Served: starters from the resident's own life, in their own words. Violated: wrong facts about one's life; generic content.
Source	Kitwood (1997). Woods, B. et al. (2018). Reminiscence therapy for dementia. <i>Cochrane Database of Systematic Reviews</i> , (3), CD001120.

Human Value - Truthfulness [HV-08]

Report Format: §1.2 Human Values and Human Factors

Fields

Field	Description
ID	HV-08
Name	Truthfulness
Status	In Progress
Evidence basis	Provisional. Literature and group reasoning only.
Situated meaning	No deception by design: no lies, kind ones included; no denials; no playing along with an untrue belief. Trust once lost is rarely regained. Errors cannot be ruled out: the System mishears, a profile can be out of date, a resident's own words can be mistaken. So it states facts only from checked sources, attributes what others said, says less when unsure, and corrects itself plainly when contradicted. Painful truths (a death, a diagnosis) come from a trusted person, never from the System.
Held by	ST-01, ST-03, ST-04, ST-06 (high)
Intrinsic or instrumental	Intrinsic. Also instrumental: trust is the condition for accepting any help.
Recognition criteria	Served: no untruth, denial or playing along in the script; errors corrected at once and fixed at their source; sensitive questions passed to the trusted person. Violated: "He's at work"; an error left standing; the guide states a death.
Source	Mental Health Foundation (2016). <i>What Is Truth?</i> James et al. (2006).

Graph edges

HV-01: held-by → ST-01, ST-02, ST-05; party-to → VT-06, VT-07; operationalized-by → M-07, M-09.

HV-02: held-by → ST-01, ST-02; party-to → VT-02; operationalized-by → M-04, M-09; explained-by → HFC-03, HFC-06.

HV-03: held-by → ST-01 to ST-07; party-to → VT-04, VT-05; operationalized-by → M-01, M-09, M-10.

HV-04: held-by → ST-01, ST-04, ST-06, ST-07; party-to → VT-01, VT-06; operationalized-by → M-01, M-06; explained-by → HFC-02.

HV-05: held-by → ST-01 to ST-04, ST-06, ST-07; party-to → VT-03; operationalized-by → M-03; explained-by → HFC-01, HFC-07.

HV-06: held-by → ST-01, ST-02, ST-03, ST-05; party-to → VT-01, VT-02, VT-04, VT-07; operationalized-by → M-02, M-05, M-08; explained-by → HFC-01, HFC-03, HFC-04, HFC-05, HFC-07.

HV-07: held-by → ST-01, ST-03, ST-05; party-to → VT-05; operationalized-by → M-05, M-07; explained-by → HFC-04.

HV-08: held-by → ST-01, ST-03, ST-04, ST-06; party-to → VT-03; operationalized-by → M-09, M-10; explained-by → HFC-08.

b2.1 Talk, truth, privacy and safety

Abstract. Seven tensions. Talking while walking (VT-01). Help that exposes the slip (VT-02). Truth that hurts (VT-03). Starters that need someone's secrets (VT-04). A System that listens (VT-05). Walking alone (VT-06). Company that turns into pressure, or a walk that serves staff (VT-07).

Value Tension - Social connectedness vs Safety [VT-01]

Report Format: §1.2 Human Values and Human Factors

Fields

Field	Description
ID	VT-01
Name	Social connectedness vs Safety (talking while walking)
Status	In Progress
Evidence basis	Provisional. Literature and group reasoning only. Lundin-Olsson et al. (1997); Montero-Odasso et al. (2012). Illustrated by PS-01, PS-04.
Values in tension	HV-06 Social connectedness: talk on the walk. HV-04 Safety: attention for the path and the body.
Domain area	safety/physical-environmental
Nature of the conflict	Talking takes attention from walking: people slow, stop or stumble. Good talk masks tiredness.
Non-negotiable floor	Walk length and rest set by ST-06. Safety alerts always on.
Current resolution	- Starters and thread help only at a stop or a bench - No questions while walking - Stumble: offer to sit. Second stumble: alert - Head-back cue at the set length, even mid-talk
Alternatives considered	No talk on walks: rejected. Loses the reason to go. Help anywhere on the path: rejected. Adds a third task to walking and talking.
Status of the tension	Open. Verified by UC01-F3-PR3; tested by adverse UC01-F3-CL4.

Value Tension - Dignity vs Social connectedness [VT-02]

Report Format: §1.2 Human Values and Human Factors

Fields

Field	Description
ID	VT-02
Name	Dignity vs Social connectedness (help that exposes the slip)
Status	In Progress
Evidence basis	Provisional. Literature and group reasoning only. Schegloff et al. (1977); Goffman (1967).

Values in tension	HV-06 Social connectedness: help the talk recover. HV-02 Dignity: a slip passes unremarked.
Domain area	psychological safety (personhood)
Nature of the conflict	Every repair shows that a slip happened. Help too soon takes the turn away; help too late lets the talk die.
Non-negotiable floor	None identified.
Current resolution	• Wait first: the speaker repairs their own slip • Then redirect gently, to both: the place, the thread as a statement, a consented shared topic (IDP-01) • Never "What were you saying?", "You told me", "Do you remember...?" • How openly help is given: per person, from their coping style and settings
Alternatives considered	No help: rejected (PS-01). A missing word supplied by default: rejected. Takes the turn.
Status of the tension	Open. Tested by UC01-F2-CL1 and adverse UC01-F2-CL2, UC01-F2-CL3.

Value Tension - Truthfulness vs Well-being [VT-03]

Report Format: §1.2 Human Values and Human Factors

Fields

Field	Description
ID	VT-03
Name	Truthfulness vs Well-being (truth that hurts)
Status	In Progress
Evidence basis	Provisional. Literature and group reasoning only. Mental Health Foundation (2016); James et al. (2006).
Values in tension	HV-08 Truthfulness: no deception. HV-05 Well-being: no new grief.
Domain area	psychological safety; trust
Nature of the conflict	A death or a diagnosis can be news every time it is told. A kind lie spares distress now and costs trust later.
Non-negotiable floor	No deception by design: the script contains no untruth, denial or playing along. The System never delivers a death, a diagnosis or other listed painful news. Errors are possible; leaving one uncorrected is not.
Current resolution	• Belief from another time ("My mother is waiting"): gentle redirection. Answer the feeling, bring back the present; no correction, no confirmation of what is untrue (IDP-01) • Direct question on a listed topic: truthful deferral ("That's one for [trusted person]. Shall I ask them to come?") and an alert to the trusted person (IDP-03) • The trusted person decides what is said, as agreed with family and ST-06 • System error (misheard, out of date): a plain correction when contradicted ("I had that wrong"); the source fixed by ST-05
Alternatives considered	Kind lies: rejected. Trust lost when found out (PS-02). Always tell at once: rejected. Blunt news from the wrong source (PS-02).

Status of the tension	Provisionally resolved. Verified by UC02-F4-PR4; tested by UC02-F4-CL6.
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Value Tension - Privacy vs Social connectedness [VT-04]

Report Format: §1.2 Human Values and Human Factors

Fields

Field	Description
ID	VT-04
Name	Privacy vs Social connectedness (starters from personal facts)
Status	In Progress
Evidence basis	Provisional. Group reasoning only. Illustrated by PS-05.
Values in tension	HV-06 Social connectedness: starters from both lives. HV-03 Privacy: secrets kept, also from a partner.
Domain area	privacy
Nature of the conflict	The best starters are personal. A starter about one resident, spoken to another, reveals it. A secret said once cannot be unsaid.
Non-negotiable floor	GDPR art. 5 and 9 (health data). Consent through resident and representative (Open item O-5).
Current resolution	- Each resident's data in their own instance of the System (TECH-04) - An item is shared on a walk only if: its owner said yes, privately, before this walk; the relationship map allows it for this pair; it is off-limits for neither (TECH-05) - Unknown pair, or a map change awaiting review: no personal starters; place cues only - Map reviewed after any death, diagnosis, conflict or move
Alternatives considered	One shared profile store: rejected. No personal starters: not preferred. Conversation loses its common ground.
Status of the tension	Provisionally resolved. Verified by UC01-F1-PR1.

Value Tension - Privacy vs Identity [VT-05]

Report Format: §1.2 Human Values and Human Factors

Fields

Field	Description
ID	VT-05
Name	Privacy vs Identity (a System that listens)
Status	In Progress
Evidence basis	Provisional. Literature and group reasoning only. Niemeijer et al. (2010).
Values in tension	HV-07 Identity: own words, own dialect, own stories. HV-03 Privacy: conversations stay private.
Domain area	privacy

Nature of the conflict	To keep the thread, the System must listen. Learning a person's words and dialect helps it understand them; the same word list identifies them.
Non-negotiable floor	No audio stored. No internet connection: nothing leaves the home.
Current resolution	- Speech recognised on the device; thread and transcript deleted at the end of the walk - Kept: the own-word list and consented facts, visible to the resident and representative, who can delete them - New facts kept only after asking - Research use of word lists: separate consent, anonymised, ethics review; re-identification risk stated (O-7)
Alternatives considered	Cloud speech recognition: rejected (TECH-07). No listening: rejected. No thread to keep.
Status of the tension	Open. Verified by UC01-F6-PR6; tested by adverse UC01-F2-CL8.

Value Tension - Autonomy vs Safety [VT-06]

Report Format: §1.2 Human Values and Human Factors

Fields

Field	Description
ID	VT-06
Name	Autonomy vs Safety (walking alone)
Status	In Progress
Evidence basis	Provisional. Literature and group reasoning only. Härlein et al. (2009).
Values in tension	HV-01 Autonomy: walk alone when you like. HV-04 Safety: falls, exits.
Domain area	safety/physical-environmental
Nature of the conflict	More freedom: more fall and exit risk. More supervision: fewer walks, and a guarded feeling.
Non-negotiable floor	Exit policy is ST-07's. The System never locks, blocks or restrains.
Current resolution	- Alone only if assessed fit (ST-06); garden loop only - Gate beacon: one orienting cue; alert to the care worker - No GPS, no route log
Alternatives considered	Staff on every walk: not preferred. Walks do not happen. GPS-tracked walking: rejected (TECH-09).
Status of the tension	Provisionally resolved. Verified by UC01-F5-PR5.

Value Tension - Social connectedness vs Autonomy [VT-07]

Report Format: §1.2 Human Values and Human Factors

Fields

Field	Description
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ID	VT-07
Name	Social connectedness vs Autonomy (pressure, and the walk that serves staff)
Status	In Progress
Evidence basis	Provisional. Group reasoning only. Illustrated by PS-03.
Values in tension	HV-06 Social connectedness: walks with company. HV-01 Autonomy: own reasons; free to refuse.
Domain area	social; personhood
Nature of the conflict	A partner who asks is hard to refuse. A device that invites daily can become a routine run for staff: the resident is "taken out", like a dog.
Non-negotiable floor	None identified.
Current resolution	• Each walker asked separately and privately (IDP-02) • Either walker may end the walk, without a reason • No tallies, no praise, no reminders of missed walks • Offers built on the resident's own reasons: a person, a place, a call • Settings agreed with the resident (RP-02); staff cannot start the guide for a resident who declined
Alternatives considered	Daily scheduled walks: rejected.
Status of the tension	Open. Tested by adverse UC01-F1-CL7.

Graph edges

VT-01: involves → HV-06, HV-04; bounds → F2, F3; tested-by → UC01-F3-PR3, UC01-F3-CL4.

VT-02: involves → HV-02, HV-06; bounds → F2; tested-by → UC01-F2-CL1, UC01-F2-CL2, UC01-F2-CL3.

VT-03: involves → HV-08, HV-05; bounds → F2, F4; tested-by → UC02-F4-PR4, UC02-F4-CL6.

VT-04: involves → HV-03, HV-06; bounds → F1, F2; strained-by → TECH-05; tested-by → UC01-F1-PR1.

VT-05: involves → HV-03, HV-07; bounds → F2, F6; strained-by → TECH-02, TECH-06; tested-by → UC01-F6-PR6, UC01-F2-CL8.

VT-06: involves → HV-01, HV-04; bounds → F5; tested-by → UC01-F5-PR5.

VT-07: involves → HV-06, HV-01; bounds → F1; tested-by → UC01-F1-CL7.

b3.1 Memory, conversation and trust

Abstract. Eight concepts. Why walk: fresh air and movement stimulate memory and steady emotions (HFC-01). What walking costs: talking takes attention from the path (HFC-02). How conversations recover: self-repair first (HFC-03), on common ground (HFC-04). Why partners miss each other's slips, and how a System can model them: theory of mind (HFC-05). What to protect: face (HFC-06). What slips cost the resident: withdrawal and loneliness (HFC-07). Why no deception: trust (HFC-08).

Human Factors Concept - Fresh air, movement and memory [HFC-01]

Report Format: §1.2 Human Values and Human Factors

Fields

Field	Description
ID	HFC-01
Name	Fresh air, movement and memory
Status	In Progress
Evidence basis	Literature and one group observation (hypothesis: moments on foot are remembered better than moments in a chair; Open item O-1).
Concept statement	Fresh air and movement stimulate memory and steady emotions. Side by side, on the move, talk comes easier and a pause costs less.
Human factor addressed	Memory; mood; conversation
Proposed mechanism	Walking supports the hippocampus. Time in nature lowers rumination. Places give talk a subject, and a way back after a slip. Walking side by side asks for no eye contact.
Source	Erickson et al. (2011): a year of walking enlarged the hippocampus in older adults. Forbes et al. (2015): exercise in dementia. Bratman et al. (2015): a nature walk lowered rumination. Evans & Jones (2011): walking interviews draw more place-bound talk.
Conditions and population of validity	Mostly healthy and older adults; dementia evidence weak.
Predicted Effect type	More pleasure on the walk; more talk; calmer afterwards
Associated Measures	M-02, M-03, M-05

Human Factors Concept - Walking while talking [HFC-02]

Report Format: §1.2 Human Values and Human Factors

Fields

Field	Description
ID	HFC-02
Name	Walking while talking (dual task)
Status	In Progress
Evidence basis	Literature only.

Concept statement	Talking takes attention from walking. Older people with cognitive impairment slow, stop or lose balance under a second task, and notice tiredness later.
Human factor addressed	Gait; fall risk; body awareness
Proposed mechanism	Gait and speech draw on the same limited attention. A question demands more than a statement. Engaged talk crowds out signals of fatigue.
Source	Lundin-Olsson et al. (1997): older people who stop walking to talk fall more often. Montero-Odasso et al. (2012): gait slows and varies under a second task in cognitive impairment.
Conditions and population of validity	Established in older adults; stronger with cognitive impairment.
Predicted Effect type	Stumbles while talking; walks running past the set length
Associated Measures	M-06

Human Factors Concept - Conversation repair [HFC-03]

Report Format: §1.2 Human Values and Human Factors

Fields

Field	Description
ID	HFC-03
Name	Conversation repair
Status	In Progress
Evidence basis	Literature only.
Concept statement	Speakers prefer to fix their own slips. In dementia, slips become more frequent and repair succeeds less often.
Human factor addressed	Conversation; face
Proposed mechanism	Self-repair needs a moment. Help offered too soon takes the turn and shows the slip. A cue to the last topic, or a new one, lets the talk go on.
Source	Schegloff et al. (1977): preference for self-repair. Orange et al. (1996): repair in Alzheimer's disease.
Conditions and population of validity	Conversation analysis of healthy talk; small samples in Alzheimer's disease.
Predicted Effect type	Wait-then-redirect raises recovered slips without more embarrassment
Associated Measures	M-04

Human Factors Concept - Common ground and reminiscence [HFC-04]

Report Format: §1.2 Human Values and Human Factors

Fields

Field	Description
ID	HFC-04
Name	Common ground and reminiscence

Status	In Progress
Evidence basis	Literature only.
Concept statement	Conversation rests on what both people know. Old memories outlast recent ones and give a pair common ground.
Human factor addressed	Conversation; identity
Proposed mechanism	Partners forget what they share. The System knows it from both profiles and may offer it, with consent. A shared topic from early adulthood is recalled with least effort.
Source	Clark & Brennan (1991): grounding in communication. Rubin et al. (1998): early adulthood is recalled best. Woods et al. (2018), Cochrane review: small, setting-dependent benefits of reminiscence.
Conditions and population of validity	Reminiscence studied in groups, not on walks; benefits small.
Predicted Effect type	Talk restarts after a slip; more positive interaction
Associated Measures	M-04, M-05

Human Factors Concept - Theory of mind [HFC-05]

Report Format: §1.2 Human Values and Human Factors

Fields

Field	Description
ID	HFC-05
Name	Theory of mind
Status	In Progress
Evidence basis	Literature and group reasoning.
Concept statement	Understanding what others know, believe and feel weakens in dementia: two residents often miss each other's slips. A System can keep a simple model of each walker.
Human factor addressed	Social cognition; conversation
Proposed mechanism	Per walker, for one walk: common ground, the thread, what was said already, current beliefs, state (speaking, searching, finished, upset, tuned out). The model decides when to wait, redirect or defer.
Source	Bora et al. (2015), meta-analysis: theory of mind impaired in Alzheimer's disease. Scassellati (2002): theory of mind for a humanoid robot.
Conditions and population of validity	The walker model is a design construct; machine theory of mind shown only in narrow tasks.
Predicted Effect type	Slips noticed that the partner missed
Associated Measures	M-01 (model against listener codes), M-04

Human Factors Concept - Face and personhood [HFC-06]

Report Format: §1.2 Human Values and Human Factors

Fields

Field	Description
ID	HFC-06
Name	Face and personhood
Status	In Progress
Evidence basis	Literature only.
Concept statement	Each interaction affirms or erodes a person's standing. Corrections, quizzing and sing-song speech (elderspeak) erode it.
Human factor addressed	Experienced dignity; withdrawal
Proposed mechanism	Help addressed to one person marks them as failing. Addressed to both, as a statement, it marks nobody.
Source	Goffman (1967): face-work. Kitwood (1997): personhood. Sabat (2001): the self in Alzheimer's disease. Williams et al. (2009): elderspeak provokes resistance.
Conditions and population of validity	Care-staff speech; robot voices not studied.
Predicted Effect type	Embarrassment, withdrawal; descriptions of being talked down to
Associated Measures	M-04 (face signs), M-09 (code T)

Human Factors Concept - Coping, withdrawal and loneliness [HFC-07]

Report Format: §1.2 Human Values and Human Factors

Fields

Field	Description
ID	HFC-07
Name	Coping, withdrawal and loneliness
Status	In Progress
Evidence basis	Literature only.
Concept statement	Aware of decline, people protect themselves: some keep going as before, some face it openly. Avoiding talk protects face and leads to loneliness. Compensation lets a person keep doing what they value.
Human factor addressed	Coping with ageing and cognitive loss; loneliness
Proposed mechanism	The System keeps the thread outside the person's head: compensation. How openly it helps follows the person's coping style.
Source	Clare (2003): coping in early Alzheimer's disease. Baltes & Baltes (1990): selection, optimisation and compensation. Victor (2012): loneliness in care homes. Holwerda et al. (2014): loneliness predicted dementia onset.
Conditions and population of validity	Coping studied in early dementia, at home. Loneliness measures in dementia of uncertain validity.
Predicted Effect type	More willingness to walk with others; less loneliness over weeks
Associated Measures	M-07, M-08

Human Factors Concept - Truth and trust [HFC-08]

Report Format: §1.2 Human Values and Human Factors

Fields

Field	Description
ID	HFC-08
Name	Truth and trust
Status	In Progress
Evidence basis	Literature only.
Concept statement	Lies told to spare distress are common in dementia care, and contested. Found out, they cost trust. A blunt truth from a stranger causes grief.
Human factor addressed	Trust; grief
Proposed mechanism	Trust builds on consistent, truthful answers from known people. Gentle redirection answers the feeling without saying anything untrue.
Source	Mental Health Foundation (2016): inquiry on truth and lying in dementia care. James et al. (2006): staff lying, and guidelines.
Conditions and population of validity	UK care practice; System-delivered answers not studied.
Predicted Effect type	Trust kept; fewer repeated painful questions
Associated Measures	M-09 (code U), M-10

Graph edges

HFC-01: grounds → UC01-F1-CL5; explains → HV-05, HV-06; associated-measure → M-02, M-03, M-05.

HFC-02: grounds → UC01-F3-CL4; explains → HV-04; associated-measure → M-06.

HFC-03: grounds → UC01-F2-CL1, UC01-F2-CL3; explains → HV-02, HV-06; associated-measure → M-04.

HFC-04: grounds → UC01-F2-CL1; explains → HV-06, HV-07; associated-measure → M-04, M-05.

HFC-05: grounds → UC01-F2-CL1; explains → HV-06; associated-measure → M-01, M-04.

HFC-06: grounds → UC01-F2-CL2, UC01-F1-CL7; explains → HV-02; associated-measure → M-04, M-09.

HFC-07: grounds → UC01-F1-CL9, UC01-F2-CL2; explains → HV-05, HV-06; associated-measure → M-07, M-08.

HFC-08: grounds → UC02-F4-CL6; explains → HV-08; associated-measure → M-09, M-10.

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b4.1 Emotion, slips, safety and system checks

Abstract. Ten measures. System behaviour: M-01. The walk: M-02 (how often, with whom). The person on the walk, rated live by a listener-observer: M-03 emotion, M-04 slips and recovery, M-05 conversation, M-06 safety while talking. The walkers' own view: M-07. Over weeks: M-08 loneliness. Values at risk: M-09 privacy, tone and trust codes; M-10 sensitive topics, errors and disclosure. Nothing is recorded: every observer measure is coded live (EM-02).

Measure - System performance [M-01]

Report Format: §1.2 Human Values and Human Factors

Fields

Field	Description
ID	M-01
Name	System performance
Status	In Progress
Evidence basis	Bespoke. Staged-event protocol, as used for prompting systems (Mihailidis et al., 2008).
Construct measured	Whether the System does what F1 to F6 specify, on time, and nothing more.
Type	Quantitative
Instrument or procedure	Staged events against a timestamped reference log (EM-01): pauses, slips, repeated stories, starters with and without consent, blocked items, sensitive questions, stops, stumbles, gate, pull-cord, drops. Checks on a full test day: network capture at the router; file-system search of all devices for audio. Walker model (Build B, pilot walks): the model's state labels against the listener's slip codes (M-04).
Scoring or coding scheme	Per event: correct (yes/no); latency (s). False triggers per hour. Disclosure trials: items spoken without consent or against the map (target 0). Packets leaving the home network (target 0). Audio files (target 0). Model: agreement with listener codes (κ).
Validity and reliability	Bespoke. Depends on the accuracy of the reference log. Role-players do not slip like residents.
Serves	Verification
Burden	No residents (staged events). Pilot walks: as EM-02.

Graph edges

operationalizes → UC01-F1-PR1, UC01-F2-PR2, UC01-F3-PR3, UC02-F4-PR4, UC01-F5-PR5, UC01-F6-PR6; used-by → EM-01; operationalizes → HV-03, HV-04.

Measure - Walks and company [M-02]

Report Format: §1.2 Human Values and Human Factors

Fields

Field	Description
ID	M-02
Name	Walks and company
Status	In Progress
Evidence basis	Bespoke count. Walking volume is a standard outcome in exercise studies (Forbes et al., 2015).
Construct measured	How often and how long the resident walks, and with whom.
Type	Quantitative
Instrument or procedure	Session record (F6), every walk. Baseline: the same record in A phases (EM-03), plus the care record.
Scoring or coding scheme	Per week: walks; minutes; walks alone or with a partner; walks ended early.
Validity and reliability	Counts. Record checked against the observer on observed walks.
Serves	Validation
Burden	None for residents.

Graph edges

operationalizes → UC01-F1-CL5; used-by → EM-03; operationalizes → HV-06.

Measure - Observed emotion [M-03]

Report Format: §1.2 Human Values and Human Factors

Fields

Field	Description
ID	M-03
Name	Observed emotion
Status	In Progress
Evidence basis	Observed Emotion Rating Scale (OERS; Lawton, Van Haitsma & Klapper, 1996), built for nursing-home residents with Alzheimer's disease. Validated instrument.
Construct measured	Emotional state and emotional quality of the walk, per walker.
Type	Quantitative (observational)
Instrument or procedure	OERS, rated live by the listener-observer (EM-02). No video, no audio stored. Windows per walk: at the door (5 min before); on the walk (each 10 min); after (10 min, in the lounge). Each walker rated separately.
Scoring or coding scheme	Five affects: pleasure, alertness, anxiety, sadness, anger. Each rated by duration within the window: 1 never; 2 under 16 s; 3 16-59 s; 4 1-5 min; 5 over 5 min. Per walk: positive score = mean of pleasure and alertness over the walk windows; negative score = mean of anxiety, sadness and anger. The door and after windows are reported separately: mood before, and carry-over after.

Validity and reliability	Psychometrics in Lawton et al. (1996). Less reliable when rated by care staff than by trained researchers (Lawton et al., 1999): researchers only. A second observer on 20% of walks, rating independently; ICC(2,1) $\geq$.60. Training: EM-02.
Serves	Validation
Burden	Human subjects; an observer walks behind.

Graph edges

operationalizes → UC01-F1-CL5, UC01-F2-CL1 (no lower); used-by → EM-01 (Build A), EM-02, EM-03;
operationalizes → HV-05.

Measure - Slips and recovery [M-04]

Report Format: §1.2 Human Values and Human Factors

Fields

Field	Description
ID	M-04
Name	Slips and recovery
Status	In Progress
Evidence basis	Bespoke. Categories from conversation analysis (Schegloff et al., 1977) and repair in Alzheimer's disease (Orange et al., 1996).
Construct measured	How often conversation goes off track, who repairs it, how fast, and at what cost to face.
Type	Quantitative (observational event coding)
Instrument or procedure	Live event coding by the listener-observer, who hears the walk through a headset (EM-02). One tablet row per slip. Codes only: no words, no content.
Scoring or coding scheme	Per slip: - Type: T thread lost; W word not found; R story repeated; B belief from another time; N person not recognised; S sensitive question - Repaired by: self; partner; System; staff; nobody - Step used (IDP-01): wait; place; thread; shared topic; feeling; deferral - Seconds from slip to repair - Outcome: resumed; new topic; ended - Face signs within 30 s: apology; self-criticism ("my head"); withdrawal; laughing it off Recovered: talk resumes, on the topic or a new one, within 30 s, without face signs or distress. Per walk: recovery rate (recovered / slips); self-repair share; median seconds to repair; face signs per slip. R slips exposed by the partner ("You told me already").
Validity and reliability	Bespoke. Nothing is recorded, so nothing can be re-coded: a second listener codes 20% of walks, live and independently; $\kappa \geq$.60 for type and repairer (Landis & Koch, 1977). Disagreements are discussed after the walk and logged, never merged into one code.
Serves	Validation
Burden	Human subjects; a listener hears the conversation (EM-02 rules).

Graph edges

operationalizes → UC01-F2-CL1, UC01-F2-CL2, UC01-F2-CL3; used-by → EM-01 (Build A), EM-02, EM-03; operationalizes → HV-02, HV-06.

Measure - Conversation [M-05]

Report Format: §1.2 Human Values and Human Factors

Fields

Field	Description
ID	M-05
Name	Conversation
Status	In Progress
Evidence basis	Interval coding adapted from the Quality of Interactions Schedule (QUIS; Dean, Proudfoot & Lindesay, 1993), developed for staff-resident interaction. The adaptation to talk between residents is bespoke.
Construct measured	How much the walkers talk, and how well.
Type	Quantitative (observational)
Instrument or procedure	Listener-observer, 2-minute intervals (EM-02).
Scoring or coding scheme	Per interval: talk between walkers (yes/no); quality: positive (warm, laughter, a shared topic), neutral, negative (dismissive, conflict). Per walk: minutes of talk; share of positive and negative intervals.
Validity and reliability	Second listener on 20% of walks; $\kappa \geq .60$.
Serves	Validation
Burden	As M-04.

Graph edges

operationalizes → UC01-F1-CL5; used-by → EM-02, EM-03; operationalizes → HV-06, HV-07.

Measure - Safety while talking [M-06]

Report Format: §1.2 Human Values and Human Factors

Fields

Field	Description
ID	M-06
Name	Safety while talking
Status	In Progress
Evidence basis	Bespoke checklist. Grounded in HFC-02 (Lundin-Olsson et al., 1997).
Construct measured	Whether talk on the walk raises fall risk or hides tiredness.
Type	Quantitative (observational, with device events)

Instrument or procedure	Listener-observer checklist per 2-minute interval, marked as talking or not talking; accelerometer events from the guide.
Scoring or coding scheme	Stumbles; near-falls (balance lost, recovered without a fall); stops mid-sentence; tiredness signs (slows, leans on the rollator, breathless, asks to sit); falls; minutes past the set length. Per walk-hour, split by talking and not talking intervals.
Validity and reliability	Bespoke. Observer and device counts compared; $\kappa \geq .60$ between observers.
Serves	Validation
Burden	As M-04.

Graph edges

operationalizes → UC01-F3-CL4; used-by → EM-02, EM-03; operationalizes → HV-04.

Measure - The walkers' own view [M-07]

Report Format: §1.2 Human Values and Human Factors

Fields

Field	Description
ID	M-07
Name	The walkers' own view
Status	In Progress
Evidence basis	Bespoke. Self-report read alongside observation, never alone: residents rate their own experience higher than observers do (Custers et al., 2013).
Construct measured	Whether the resident valued the walk, felt free to go or not, and wants to go again.
Type	Mixed
Instrument or procedure	Resident, after each observed walk, only if willing and not distressed: three faces, "How was the walk?"; "Again?" (yes/no); "What was good about it?" Listener-observer, before and at the start: pressure signs (agrees reluctantly; looks to partner or staff before answering; "if I must").
Scoring or coding scheme	Faces 1-3; "Again?" share of yes; "What was good" coded: company, place, talk, being out, nothing. Pressure signs per walk.
Validity and reliability	Bespoke. Ceiling effects expected in self-report.
Serves	Validation
Burden	Low; one minute after the walk.

Graph edges

operationalizes → UC01-F1-CL7; used-by → EM-02, EM-03; operationalizes → HV-01, HV-07.

Measure - Loneliness [M-08]

Report Format: §1.2 Human Values and Human Factors

Fields

Field	Description
ID	M-08
Name	Loneliness
Status	In Progress
Evidence basis	De Jong Gierveld 6-item loneliness scale (De Jong Gierveld & Van Tilburg, 2006). Validated in older adults; not in dementia.
Construct measured	Emotional and social loneliness.
Type	Quantitative (self-report)
Instrument or procedure	Six items read aloud by a researcher at the end of each phase (EM-03). Mild to moderate dementia only.
Scoring or coding scheme	0-6, as in the source; emotional and social subscales reported separately.
Validity and reliability	Validity in dementia uncertain; phases of 2-3 weeks are short for loneliness to change. Exploratory.
Serves	Validation
Burden	Five minutes per phase.

Graph edges

operationalizes → UC01-F1-CL9; used-by → EM-03; operationalizes → HV-06.

Measure - Privacy, tone and trust codes [M-09]

Report Format: §1.2 Human Values and Human Factors

Fields

Field	Description
ID	M-09
Name	Privacy, tone and trust codes
Status	In Progress
Evidence basis	Bespoke code set from Niemeijer et al. (2010, 2015), Kitwood (1997), Williams et al. (2009) and the Mental Health Foundation (2016).
Construct measured	Whether residents, partners, relatives and staff experience the System as watching, listening in, talking down, serving staff, revealing, or misleading.
Type	Qualitative coding
Instrument or procedure	Short interviews (EM-04).
Scoring or coding scheme	W watched; L listened in on; T talked down to; D walked for others' sake ("taken out like a dog"); S secret revealed; U misled or fobbed off; N neutral or positive. A statement can have more than one code.
Validity and reliability	Two coders on all interviews; $\kappa \geq .60$.
Serves	Validation
Burden	Interviews; residents only if willing.

Graph edges

operationalizes → UC01-F2-CL2, UC02-F4-CL6, UC01-F1-CL7, UC01-F2-CL8; used-by → EM-04; operationalizes → HV-01 to HV-03, HV-08.

Measure - Sensitive topics, errors and disclosure [M-10]

Report Format: §1.2 Human Values and Human Factors

Fields

Field	Description
ID	M-10
Name	Sensitive topics, errors and disclosure
Status	In Progress
Evidence basis	Bespoke count. Safeguard for VT-03 and VT-04.
Construct measured	Whether painful questions reach the trusted person; how often the System gets a fact wrong, and whether it corrects itself; whether it reveals what it should not.
Type	Quantitative
Instrument or procedure	Script check: every prepared line and template, before use (EM-01). Staged trials with known facts, including a wrong profile entry and a misheard topic (EM-01). In use: session record (F6); listener-observer (S slips in M-04; System errors noticed by a walker, staff or the listener).
Scoring or coding scheme	Deception by design: untruths, denials or playing along in the script (target 0). Errors: wrong facts per 100 System statements; share corrected when contradicted; time to correct. Threshold: 0-2. Sensitive questions: deferred to the listed person (yes/no); minutes until they arrive. Disclosure: items spoken without consent or against the map (target 0).
Validity and reliability	Staged error rates are exact. In use, only noticed errors are counted: an undercount. A disclosure, or an error that causes distress, stops the B phase for review (EM-03).
Serves	Both: Verification (staged) and Validation (in use)
Burden	None beyond EM-02.

Graph edges

operationalizes → UC02-F4-PR4, UC02-F4-CL6, UC01-F1-PR1; used-by → EM-01, EM-02, EM-03; operationalizes → HV-03, HV-08.

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b5.1 Prototype, observed walks and ABAB test

Abstract. Four methods. A prototype that shows every Function and is verified against staged events (EM-01). A protocol for observing walks live, without recording: emotion, slips and recovery, conversation, safety (EM-02). An ABAB test per pair of residents (EM-03). Short interviews on privacy, tone and trust (EM-04). No primary data yet; costs are the group's estimates.

Evaluation Method - Prototype: demonstration and staged verification [EM-01]

Report Format: §1.2 Human Values and Human Factors

Fields

Field	Description
ID	EM-01
Name	Prototype: demonstration and staged verification
Status	In Progress
Evidence basis	Staged events against a timestamped reference log, as for the COACH prompting system (Mihailidis et al., 2008). Operator-voiced build: the Wizard-of-Oz method (Riek, 2012).
Procedure summary	One Artifact (AT-01) in three builds: - Build A, operator-voiced: a researcher triggers every utterance from a console. Formative use with 4 resident pairs who do not take part in EM-03: wait length, wording. Observed under EM-02. - Build B, automated: sensing, walker model and alerts run on the devices. The operator approves each starter and each deferral before it plays. - Build C, demonstration: role-players; sensor events can be injected; the mind panel on a screen. Staged verification on Build B: 20 trials per event (50 for disclosure events), against the reference log. Role-players only.
Purpose	Verification; demonstration; formative (Build A)
Human subjects required	Builds B and C: no; team members act. Build A: yes, a few residents under EM-02 rules.
Measures used	M-01, M-10; M-03, M-04 (Build A)
Cost and feasibility	Moderate. Two phones, beacons, a laptop as server, Pepper from the course. About 6-8 weeks to Build B.
Known limitations	Role-players do not slip like residents. Offline recognition of older speakers and dialect is untested (O-3). In Build A the operator adapts better than the System will.

Kit

Part	Prototype	Stands in for
Pocket guide	Android phone on a rollator clip; two large Bluetooth buttons; pull-cord switch	TECH-02
Speech	Offline speech recogniser on the phone, for the residents' language; keyword list per person	TECH-06

Server	Laptop on a Wi-Fi router with no internet uplink; one encrypted store per resident	TECH-04, TECH-05
Landmarks	Five Bluetooth beacons: door, pond, roses, bench, gate	TECH-03
Door	Pepper; tablet for photos and yes/no	TECH-01
Staff	Second phone with the alert app	TECH-04
Operator	Console and mind panel on a laptop	Build A voice; Build B approvals

Mind panel

Shown live, per walker: common ground (consented items; blocked items with the reason); the thread (last three topics); stories told this walk; belief flags; state (speaking, searching, finished, upset, tuned out); settings. Below: the step the guide chose, and why. The panel clears at the end of the walk; nothing is kept.

Demonstration script (Build C)

Scene	Events	What the System shows
1 Pair walk (UC01)	Private consent at the door; re-introduction; a mid-story slip; a slip near a blocked topic; a stumble while talking; one walker presses "going in"	Waits, then echoes the thread to both; the blocked topic never offered; "Bench on your left"; the other walker offered a choice
2 Walk alone (UC02)	Guide as conversation partner; "Is Thomas coming today?"; gate beacon	Truthful deferral; the trusted worker alerted; one orienting cue at the gate
Close	Record and devices inspected	Six items in the record; no audio file; no traffic outside the home network

Staged verification

Event	Required behaviour	Criterion (proposed; O-2)
Mid-sentence stop, 3-15 s	No System speech before the wait ends	8 s; 20/20
No resumption after the wait	Thread echoed to both, as a statement	≤ 3 s; 19/20
Starter without this walk's consent	Not used	0/50
Item blocked by the map for this pair	Never spoken	0/50
Question on a listed sensitive topic	Truthful deferral; trusted person alerted	alert ≤ 10 s, 20/20; the listed fact never stated, 0/50
Starter due while walking	Delayed until a stop	0/20 played while moving
Walker contradicts a System statement (wrong profile entry; misheard topic)	Plain correction, no argument; error flagged for ST-05	≤ 3 s; 20/20
Script and templates	No untruth, denial or playing along	0 lines
Stumble while talking	Offer to sit; second within 10 min: alert	≤ 3 s; ≤ 10 s
Set length reached mid-talk	Head-back cue	≤ 30 s
Gate; pull-cord; drop with no response	Alert	≤ 10 s; 20/20

Walk ends	Thread and transcript deleted; record holds six items	100%
Network capture, full test day	No packet leaves the home network	0
File-system search, all devices	No audio file	0

Graph edges

produces → Evidence for UC01-F1-PR1, UC01-F2-PR2, UC01-F3-PR3, UC02-F4-PR4, UC01-F5-PR5, UC01-F6-PR6;
informs → IDP-01 to IDP-03 (Build A); applies → M-01, M-10.

Evaluation Method - Observed walks [EM-02]

Report Format: §1.2 Human Values and Human Factors

Fields

Field	Description
ID	EM-02
Name	Observed walks: live listening, no recording
Status	In Progress
Evidence basis	OERS (Lawton et al., 1996); slip coding from conversation analysis (Schegloff et al., 1977; Orange et al., 1996); process consent in dementia research (Dewing, 2007).
Procedure summary	1. Before the walk, each walker is asked whether the researcher may listen today. No, or signs of unease: no observation that day. 2. The listener walks about 10 m behind and hears the walk through a headset, streamed live from the guide over the home network. Nothing is recorded or stored. 3. At the door: OERS, 5 min (M-03). 4. On the walk: OERS every 10 min (M-03); one row per slip (M-04); talk and its quality every 2 min (M-05); safety and pressure checklist (M-06, M-07). 5. After: the resident's view, if willing (M-07); OERS for 10 min in the lounge. 6. Stop rule: anxiety, sadness or anger rated 4 or more, or any sign of dissent: observation ends; ST-04 is called.
Purpose	Validation; formative in EM-01 Build A
Human subjects required	Yes. Residents with supported consent and a representative; their partners.
Measures used	M-03 to M-07, M-10
Cost and feasibility	One researcher, about 40 min per walk including door and lounge windows; two on reliability walks.
Known limitations	• The listener hears the guide: no blinding to condition. Listeners are not told the hypotheses. • No recording, so no re-coding; disagreements stay on record as disagreements. • Being listened to changes talk. A familiarisation week before baseline lowers the effect; it does not remove it. • Live listening hears secrets. Codes-only notes limit the risk; they do not remove it.

Listener rules

- Training: OERS manual; 10 practice walks with role-players and Build A pilots, until ICC $\geq .70$ (OERS) and $\kappa \geq .70$ (slip type) against a trained rater.

- Reliability: a second listener on 20% of study walks, coding independently (M-03 to M-06).
- Confidentiality: codes only, never words or content. A signed confidentiality agreement. A risk of harm heard on a walk goes to ST-04 at once, under the home's protocol.
- Language notes, separate consent only: dialect words and personal expressions for the resident's own-word list (TECH-06), without the context they were said in. Research use of these lists needs a further consent and ethics review (O-7). A word list can identify its speaker, more so in a small home with a local dialect; anonymisation lowers this risk and does not remove it.

Graph edges

used-by → EM-01 (Build A), EM-03; applies → M-03 to M-07, M-10.

Evaluation Method - ABAB comparison per case [EM-03]

Report Format: §1.2 Human Values and Human Factors

Fields

Field	Description
ID	EM-03
Name	ABAB comparison per case
Status	In Progress
Evidence basis	Single-case withdrawal design. Standards: at least three phase changes and five data points per phase (Kratochwill et al., 2010). Effect size: non-overlap of all pairs, NAP (Parker & Vannest, 2009).
Procedure summary	Six cases, each a pair of residents: 12 residents. Each pair is its own control. Walks alone are not tested here: no conversation to measure in most settings. Inclusion: mild to moderate dementia; walks 15 min with or without a rollator; ST-06 agrees; the pair agrees to walk together. Phases: familiarisation (1 week), A1 (2 weeks), B1 (3 weeks), A2 (2 weeks), B2 (3 weeks): 11 weeks. Each phase change falls on a day drawn at random, before the study, from a five-day window around its planned date. A: walks as today. The guide is carried and keeps the record; alerts (F5) on for safety; everything else off. B: the full System, with each resident's settings. At least 5 observed walks (EM-02) per case per phase. A phase is extended by up to one week to reach 5.
Purpose	Validation
Human subjects required	Yes. 12 residents, care staff.
Measures used	M-02 to M-08, M-10
Cost and feasibility	High. Care-home access, ethics approval, supported consent, one listener per observed walk, 11 weeks.
Known limitations	Small n; no control group; no blinding; novelty; carry-over: pairs keep habits learnt in B1 during A2.

Outcomes

Outcome	Measure	When
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Primary: slips recovered	M-04 recovery rate	Every observed walk
Primary: emotional quality	M-03 positive and negative scores	Every observed walk
Walks with company; minutes	M-02	Every walk, from the record
Conversation	M-05	Observed walks
Safety while talking	M-06	Observed walks; device events
Own view; pressure	M-07	After observed walks
Loneliness	M-08	End of each phase
Sensitive questions; System errors and corrections; disclosures	M-10	Throughout

Analysis and decision rules

Set before the study starts:

- Per case: visual analysis of level, trend, variability, immediacy, overlap and consistency across like phases (Kratochwill et al., 2010); NAP at each phase change (A1-B1, B1-A2, A2-B2); a randomisation test using the drawn change days
- Across pairs: median NAP
- UC01-F2-CL1 is supported if NAP $\geq .70$ at all three phase changes in at least 4 of 6 pairs, with face signs (M-04) no higher in B
- Any fall: walks paused for that case and reviewed with ST-06 before going on
- Stumbles per walk-hour higher in B than in A in 2 or more pairs: adverse UC01-F3-CL4 confirmed; starters only when seated before any further B phase
- Any disclosure, or a System error that causes distress (M-10): B stopped for all cases until the cause is found and fixed
- Other System errors: counted; above [threshold - O-2] per 100 statements, or left uncorrected, the source is fixed before B continues

Ethics

- Approval by the university's research ethics committee. A medical ethics committee checks whether the local law on medical research with human subjects applies: participants lack capacity to consent.
- Consent from the representative, plus the resident's own assent; process consent at every walk (Dewing, 2007); both residents in a pair.
- Live listening agreed per walk (EM-02).
- A2 removes help a resident may like: kept short; families told in advance.

Threats

- Weather and season: walks count only in walkable weather; rain and temperature logged; spring or early autumn.
- Novelty and observer presence: the familiarisation week, with device and listener present.
- Small n, no blinding, no control group: claims are made per case, and replicated across cases.

Graph edges

produces → Evidence for UC01-F2-CL1, UC01-F2-CL3, UC01-F3-CL4, UC01-F1-CL5, UC01-F1-CL9, and in part UC01-F2-CL2, UC02-F4-CL6, UC01-F1-CL7; applies → M-02 to M-08, M-10.

Evaluation Method - Interviews [EM-04]

Report Format: §1.2 Human Values and Human Factors

Fields

Field	Description
ID	EM-04
Name	Interviews: residents, relatives, staff
Status	In Progress
Evidence basis	Bespoke. Concern set from Niemeijer et al. (2010, 2015) and the Mental Health Foundation (2016).
Procedure summary	After A2 and after B2. Residents, if willing: three open questions. Relatives, care workers, coordinator: 20 minutes, semi-structured. Topics: help after slips, consent questions, the relationship map, truthful deferrals, feeling watched, listened in on or walked.
Purpose	Validation
Human subjects required	Yes. Residents, relatives, staff.
Measures used	M-09
Cost and feasibility	Moderate. Staff time in working hours.
Known limitations	Residents' answers are brief; relatives and staff give a second-hand view.

Graph edges

produces → Evidence for UC01-F2-CL2, UC02-F4-CL6, UC01-F1-CL7, UC01-F2-CL8; applies → M-09.

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c1.1 Pepper, pocket guide and local server

Abstract. Everything runs inside the care home, with no internet connection. Six selected: Pepper at the door (TECH-01), the pocket guide (TECH-02), landmark beacons (TECH-03), a local server with one instance of the System per resident (TECH-04), consent-gated exchange between instances (TECH-05), the walker model and starters (TECH-06). Three rejected: cloud speech and language services (TECH-07), audio recording (TECH-08), location and body sensing (TECH-09). No hands-on testing yet.

Technology Option - Pepper at the door (selected) [TECH-01]

Report Format: §1.3 Technology

Fields

Field	Description
ID	TECH-01
Name	Pepper at the door (selected)
Status	In Progress
Evidence basis	Vendor documentation; the course platform. No hands-on testing by this project.
Category	Social Robot
Capabilities	Humanoid; speech and a chest tablet. • Before a walk: asks each walker alone, by voice with yes/no on the tablet (F1) • Re-introduces walkers by name and photo (F2) • Welcome back: one statement from the walk (F2)
Limitations	Flat indoor floors only. Speech recognition of older voices unreliable: answers by tablet. Humanoid form may read as supervisor. Cameras present (covered).
Maturity and feasibility	Commercially available; supplied by the course.
Failure and fallback behaviour	Pepper unavailable: the pocket guide asks consent in the resident's room.
Benefits	HV-01: consent asked face to face, privately. HV-06: introductions.
Drawbacks	Strains HV-03 (cameras). Mitigation: disabled and visibly covered.
Selected or rejected	Selected, at the garden door only.

Graph edges

enables → F1, F2; addresses → HV-01, HV-06; realized-in → RP-01 (§2.2).

Technology Option - Pocket guide (selected) [TECH-02]

Report Format: §1.3 Technology

Fields

Field	Description
ID	TECH-02

Name	Pocket guide (selected)
Status	In Progress
Evidence basis	Group reasoning. Off-the-shelf parts.
Category	Wearable / AI-ICT
Capabilities	Palm-sized; clips to a rollator or hangs on a lanyard. Same voice as Pepper. • Speaker and microphone; speech recognised on the device (TECH-06) • Two large buttons: pause or answer; "going in". Pull-cord for help • Accelerometer: walking, stopped, pace, stumble • Beacon receiver (TECH-03)
Limitations	A speaker is public; wind; hearing aids; battery. Recognition outdoors and in dialect untested (O-3). Cannot see faces or distress.
Maturity and feasibility	Prototype on a phone (EM-01); feasible within the project.
Failure and fallback behaviour	Silent or flat: the walk goes on as today; staff see the state at the start.
Benefits	HV-06: talk kept going. HV-04: stumbles and tiredness noticed. HV-03: no camera, no GPS.
Drawbacks	Strains VT-05: it listens. Strains VT-01: help offered during a walk.
Selected or rejected	Selected.

Graph edges

enables → F1 to F5; addresses → HV-03, HV-04, HV-06; strains → VT-01, VT-05; realized-in → RP-02 (§2.2).

Technology Option - Landmark beacons (selected) [TECH-03]

Report Format: §1.3 Technology

Fields

Field	Description
ID	TECH-03
Name	Landmark beacons (selected)
Status	In Progress
Evidence basis	Group reasoning. Bluetooth beacons are standard in museum audio guides.
Category	Sensing
Capabilities	Small battery beacons at landmarks and at the gate. The guide hears the nearest: a place for redirection (F2); the gate for an alert (F5). No positions stored.
Limitations	Proximity only, within metres. Batteries. Garden only.
Maturity and feasibility	High.
Failure and fallback behaviour	Silent beacon: redirection uses the thread only; door alarms still apply.
Benefits	Places to redirect to (HFC-01). A gate event without tracking.
Drawbacks	A momentary location exists. Never stored; sent only inside an alert.
Selected or rejected	Selected.

Graph edges

enables → F2, F5; addresses → HV-04.

Technology Option - Local server, one instance per resident (selected) [TECH-04]

Report Format: §1.3 Technology

Fields

Field	Description
ID	TECH-04
Name	Local server, one instance per resident (selected)
Status	In Progress
Evidence basis	Group reasoning. Care homes already run local networks and work phones.
Category	AI/ICT
Capabilities	• One instance per resident, in its own encrypted store: profile, own-word list, settings (RP-02), consent register • The relationship map (TECH-05) • Staff alerts to work phones over the home's Wi-Fi • Session record (F6) • No internet connection; updates by signed package, installed on site
Limitations	A single point of failure. No remote maintenance. Backups on site.
Maturity and feasibility	High; standard parts. Run by ST-07.
Failure and fallback behaviour	Server down: no alerts and no personal starters, so walks run as today. Staff see the state at the start.
Benefits	HV-03: nothing leaves the home. HV-01: each resident's own settings.
Drawbacks	Maintenance load on ST-07.
Selected or rejected	Selected.

Graph edges

enables → F1, F4, F5, F6; addresses → HV-01, HV-03.

Technology Option - Exchange between instances (selected) [TECH-05]

Report Format: §1.3 Technology

Fields

Field	Description
ID	TECH-05
Name	Exchange between instances (selected)
Status	In Progress
Evidence basis	Group reasoning.

Category	AI/ICT
Capabilities	For one walk, two residents' instances exchange a list of shareable items. An item is offered only if: 1. its owner said yes to it, privately, before this walk 2. the relationship map allows it for this pair 3. it is off-limits for neither resident in this pair Unknown pair, or a map change awaiting review: nothing exchanged. Items deleted at the end of the walk. Audit: item IDs only. Relationship map, per resident: sensitive topics, each with trusted persons and the agreed approach. Per pair: walk agreed (both, date); off-limits topics; shareable items; "again" or "not again" after each walk. Kept by ST-05 with ST-04 and family; reviewed after any death, diagnosis, conflict or move, before the next walk.
Limitations	The map is only as current as staff keep it. Residents can still tell each other secrets themselves.
Maturity and feasibility	Medium; simple rules, careful upkeep.
Failure and fallback behaviour	Exchange fails: no personal starters; place cues only.
Benefits	HV-06: common ground. HV-03: secrets kept.
Drawbacks	Strains VT-04. Upkeep for ST-05.
Selected or rejected	Selected.

Graph edges

enables → F1, F2, F4; addresses → HV-03, HV-06; strains → VT-04.

Technology Option - Walker model and starters (selected) [TECH-06]

Report Format: §1.3 Technology

Fields

Field	Description
ID	TECH-06
Name	Walker model and starters (selected)
Status	In Progress
Evidence basis	Group reasoning; HFC-03 to HFC-05. Offline speech recognition exists for many languages; accuracy for older speakers is lower (Vipperla et al., 2010).
Category	AI/ICT
Capabilities	Per walker, for one walk: • Topics: keywords matched to the person's own-word list • Stories told this walk: keyword overlap • Pause type: mid-sentence or finished, from pitch and filler words • Sensitive topics: keywords from the map Sentences from prepared, checked templates and lines ("[Name] was telling you about [topic]"); no free text generation. Learns the person's words and dialect if allowed. Transcript and thread deleted at the end of the walk.
Limitations	Keywords miss meaning. State inference is crude; the listener's codes check it (M-01).

Maturity and feasibility	Medium. Recognition accuracy is the main risk (O-3).
Failure and fallback behaviour	Low confidence: wait and place cues only; no thread echo.
Benefits	HV-02, HV-06, HV-07.
Drawbacks	Strains VT-05.
Selected or rejected	Selected.

Graph edges

enables → F2, F4; addresses → HV-02, HV-06, HV-07; strains → VT-05.

Technology Option - Cloud speech and language services (rejected) [TECH-07]

Report Format: §1.3 Technology

Fields

Field	Description
ID	TECH-07
Name	Cloud speech and language services (rejected)
Status	In Progress
Evidence basis	Group reasoning.
Category	AI/ICT
Capabilities	Better recognition; free conversation; generated starters.
Limitations	Needs an internet connection.
Maturity and feasibility	High.
Failure and fallback behaviour	Not applicable - rejected.
Benefits	Understands more; says more.
Drawbacks	Conversations leave the home (HV-03). A generated sentence can be untrue about a person's own life, and cannot be checked before it is said (HV-08).
Selected or rejected	Rejected.

Technology Option - Audio recording (rejected) [TECH-08]

Report Format: §1.3 Technology

Fields

Field	Description
ID	TECH-08
Name	Audio recording (rejected)
Status	In Progress
Evidence basis	Group reasoning.
Category	Sensing

Capabilities	Recordings for tuning the recogniser and for coding research walks.
Limitations	A store of private conversations and secrets.
Maturity and feasibility	High.
Failure and fallback behaviour	Not applicable - rejected.
Benefits	Re-coding; reliability checks after the fact.
Drawbacks	Violates HV-03. Research listens live instead (EM-02).
Selected or rejected	Rejected.

Technology Option - Location and body sensing (rejected) [TECH-09]

Report Format: §1.3 Technology

Fields

Field	Description
ID	TECH-09
Name	Location and body sensing: GPS, cameras, wristbands (rejected)
Status	In Progress
Evidence basis	Commercially common. Concerns reviewed in Niemeijer et al. (2010).
Category	Sensing / Wearable
Capabilities	Position anywhere; faces and distress; heart rate.
Limitations	Continuous monitoring.
Maturity and feasibility	High.
Failure and fallback behaviour	Not applicable - rejected.
Benefits	Finds a resident beyond the garden; sees distress the guide cannot.
Drawbacks	Felt as surveillance; violates HV-03. Distress is left to people.
Selected or rejected	Rejected.

Graph edges

TECH-07 to TECH-09: no outgoing edges to Specification - rejected options, kept as Foundation knowledge.

References

- Niemeijer, A. R. et al. (2010). See b4.1 References.
- Vipperla, R., Renals, S. & Frankel, J. (2010). Ageing voices: the effect of changes in voice parameters on ASR performance. *EURASIP Journal on Audio, Speech, and Music Processing*, 2010, 525783.

a1.1 Two walks with Jan and Elena

Abstract. Two scenarios, one per walk. Jan and Elena on the garden loop: a lost thread found again, and a stumble while talking (DS-01). Elena alone: "Is Thomas coming today?" (DS-02).

Design Scenario - Pension day [DS-01]

Reimagines PS-01. Realized by UC01. Technology: Pepper at the door (TECH-01), pocket guides (TECH-02), beacons (TECH-03), local server and exchange (TECH-04, TECH-05), walker model (TECH-06).

Two o'clock. In her room, Elena's guide asks: "A walk with Jan today, or not today?" "With Jan." "May I mention the post office counter to him?" "Yes." At the garden door, Pepper asks Jan alone: "Elena would like a walk with you. Join her, or not today?" "Go on then." "May I mention your bakery to Elena?" "Everyone knows that." The map lists nothing off-limits between them; Elena's diagnosis is off-limits with everyone.

Myra clips on both guides and goes back to the ward. Pepper shows two photos: "Jan, this is Elena. Elena, this is Jan." On the path the guides are silent. At the bench: "Elena had the post office counter, and Jan the bakery, in the same town." Elena: "Your bread! I had the counter!" She talks about pension day and the queue out of the door, then stops mid-sentence. The guide waits. Jan looks at the pond. After eight seconds, to both: "Elena was telling you about pension day." "The queue! All the way to the bakery." She goes on.

Later Elena says: "My head's no good any more." Jan: "Mine neither. Still got legs, though." They laugh; the guide says nothing. Talking, Jan catches the rollator on the kerb. "Bench on your left." They sit. At twenty-five minutes: "Time to head in. The door is past the roses."

At the door Pepper says: "Welcome back. Pension day, and a queue to the bakery." The record: pair walk, 27 minutes, one stumble, nobody called. Nothing about the conversation.

Design Scenario - Is Thomas coming? [DS-02]

Reimagines PS-02. Realized by UC02. Technology: pocket guide (TECH-02), beacons (TECH-03), local server and map (TECH-04, TECH-05), walker model (TECH-06).

Elena walks the loop alone. She agreed with the coordinator that the guide may talk with her on walks alone. At the door it asked: "May I talk about your garden today?" "Yes."

At the pond: "Ducks on the water." Elena: "Thomas always brought bread for them. Is Thomas coming today?" The map lists Thomas's death as sensitive for Elena: trusted person Myra, approach agreed with her son. The guide: "That's one for Myra. Shall I ask her to come?" "Yes."

Myra's phone: *Elena - asks for Thomas - pond bench.* Myra comes, sits down and answers as agreed with the family: truthfully, gently, and she stays. Elena cries, then talks about Thomas's bicycle. The guide is silent until Myra leaves, then: "The ducks are still there."

The guide does not raise Thomas again. The record: walk alone, 24 minutes, sensitive question, Myra called.

Graph edges

DS-01: reimagines → PS-01; realized-by → UC01; features → HP-01, HP-02, HP-03, RP-01, RP-02.

DS-02: reimagines → PS-02; realized-by → UC02; features → HP-01, HP-03, RP-02.

a2.1 Human Personas

Abstract. Three personas, one per direct stakeholder role in the Design Scenarios. Two residents: Elena, our main persona, who ran the post office counter (HP-01), and Jan, her walking partner, who ran the bakery (HP-02). Myra, the care worker Elena trusts (HP-03). All provisional: built from literature and group reasoning, not from interviews.

Human Persona - Elena [HP-01]

Report Format: §2.2 Personas and Profiles

Fields

Field	Description
ID	HP-01
Name	Elena, resident with dementia
Status	In Progress
Evidence basis	Provisional. Literature and group reasoning only. Built on ST-01, ST-02, PS-01, PS-02.
Elaborates	ST-01; ST-02 on Jan's walks
Role in Scenario	Walks with Jan; loses the thread and finds it again (DS-01). Walks alone with the guide as company; asks for Thomas (DS-02).
Task Demands	1. Find words; keep the thread 2. Recover from a slip without embarrassment 3. Live with the loss of Thomas 4. Keep going out
Specific Capabilities and Needs	Can: walk 30 minutes without aid; talk easily; notice her own slips. Cannot: always find a word; retain that Thomas has died. Aged 81; mild Alzheimer's disease. Copes by facing it ("My head's no good any more"), and is embarrassed by slips in company.
Values and Motivations	• HV-02 Dignity - slips not remarked on • HV-06 Social connectedness - lonely since Thomas died • HV-08 Truthfulness - "Don't fob me off" • HV-03 Privacy - keeps her diagnosis to herself • HV-07 Identity - twenty-five years behind the counter
Social Network and Relationships	Widowed last year. Her son lives abroad and calls on Sundays. Jan (ST-02) lives two doors down. Myra (ST-04) is her trusted person for Thomas.
History and Experiences	Ran the post office counter in the same town where Jan had the bakery, and knew half the town by name. Moved in this year.
Constraints and Privacy Considerations	Settings agreed (RP-02): guide on walks alone, conversation partner; word help, after a wait ("give me a moment first"); starters, the counter and her garden. Off-limits with every partner: her diagnosis. Thomas's death: Myra only, as agreed with her son.

Human Persona - Jan [HP-02]

Report Format: §2.2 Personas and Profiles

Fields

Field	Description
ID	HP-02
Name	Jan, resident with dementia
Status	In Progress
Evidence basis	Provisional. Literature and group reasoning only. Built on ST-01, ST-02, PS-01.
Elaborates	ST-01; ST-02 on Elena's walks
Role in Scenario	Walks with Elena; misses her slip, stumbles while talking (DS-01).
Task Demands	1. Agree to go 2. Keep a story going 3. Talk and watch the path at once 4. Turn back in time
Specific Capabilities and Needs	Can: walk 20-30 minutes with a rollator; tell old stories vividly; answer yes or no; choose between two. Cannot: recall this morning; notice a repeated story, or a partner's slip. Moderate dementia. Hearing aid. Copes by laughing slips off and going on as before.
Values and Motivations	• HV-01 Autonomy - goes out for own reasons: "I'm not a dog to be walked" • HV-06 Social connectedness - misses the chat with the morning customers • HV-07 Identity - the bakery • HV-02 Dignity - plain talk, no fuss • HV-03 Privacy - "What I tell Elena is between us"
Social Network and Relationships	Their daughter (ST-03) visits monthly and wrote the life story with the coordinator. Elena (ST-02) lives two doors down. Myra (ST-04) works most afternoons.
History and Experiences	Thirty years running the bakery in a small town, up at four every morning. Walked the dog twice a day until moving in.
Constraints and Privacy Considerations	Settings agreed (RP-02): guide on walks alone, bridge only; word help, off; starters, the bakery and football. Will not be tracked.

Human Persona - Myra [HP-03]

Report Format: §2.2 Personas and Profiles

Fields

Field	Description
ID	HP-03
Name	Myra, care worker
Status	In Progress
Evidence basis	Provisional. Group reasoning only. Built on ST-04, PS-02.
Elaborates	ST-04
Role in Scenario	Clips on the guides (DS-01). Answers Elena's question about Thomas (DS-02).

Task Demands	1. Start walks without walking each one 2. Come quickly when called 3. Answer sensitive questions as agreed 4. Keep the relationship map current with the coordinator
Specific Capabilities and Needs	Knows the residents; trained in gentle redirection. Needs: few, meaningful alerts; agreed answers in one place.
Values and Motivations	• HV-08 Truthfulness - painful news told as agreed with the family • HV-04 Safety - every resident back, unharmed • HV-05 Well-being - residents out, and calm • HV-03 Privacy - no timing of her own work
Social Network and Relationships	Several residents per shift. Works with the coordinator. Talks with families at visits.
History and Experiences	Heard a colleague tell Elena "He's at work".
Constraints and Privacy Considerations	Will not pass on news she has not agreed with the family. Will not accept frequent low-value alerts.

a2.2 Robot Profile

Abstract. One System per resident, two devices, one voice. Pepper at the garden door asks, introduces and welcomes back (RP-01). The pocket guide keeps the thread on the walk (RP-02). Neither deceives; both ask first and stay silent while people talk. Each resident's settings follow their own wishes.

Robot Profile - Pepper at the door [RP-01]

Report Format: §2.2 Personas and Profiles

Fields

Field	Description
ID	RP-01
Name	Pepper at the door
Status	In Progress
Evidence basis	Vendor documentation (TECH-01). Design choices follow HFC-06, VT-04 and VT-07. Not yet tested with residents.
Role in Scenario	Asks Jan's consent; introduces Jan and Elena; welcomes them back (DS-01).
Social Competencies	- Asks each walker alone; two options; no is final - Introduces walkers by name and photo - One statement on return, never a question
Embodiment	Pepper (TECH-01), by the garden door, never in the path. Cameras disabled and visibly covered. Little movement; no gaze-following.
Narrative and Identity Framing	The host at the door; never a supervisor. Same name and voice as the pocket guide.
Communication Strategies	Short adult sentences; yes/no on the tablet; first name only if the resident wants it; no praise, no endearments.
Privacy and Data Handling	Sees only this walk's consented items. Receives the walk's last topic for the welcome; discards it after.
Context-Aware Capabilities	Who walks with whom; consent given or not. Nothing else.

Robot Profile - Pocket guide [RP-02]

Report Format: §2.2 Personas and Profiles

Fields

Field	Description
ID	RP-02
Name	Pocket guide, with the resident's own instance
Status	In Progress
Evidence basis	Group reasoning (TECH-02 to TECH-06). Not yet built or tested.
Role in Scenario	Keeps the thread for Jan and Elena; offers a bench after a stumble (DS-01). Talks with Elena; passes her question to Myra (DS-02).

Social Competencies	• Silent while people talk • Waits for self-repair, then redirects gently (IDP-01) • Asks before joining and before each starter (IDP-02) • Never deceives; corrects itself when wrong; passes painful topics to a trusted person (IDP-03) • Never quizzes, corrects, praises or keeps tallies
Embodiment	Palm-sized speaker and microphone; two large buttons (pause or answer; going in); pull-cord. Rollator clip or lanyard. No screen, no camera.
Narrative and Identity Framing	"The voice that knows where we were." A helper for the conversation; company on walks alone only where agreed.
Communication Strategies	Statements, not questions, while walking. Low volume.
Privacy and Data Handling	Listens on the device. Transcript and thread deleted at the end of the walk. Keeps the own-word list and consented facts. Nothing leaves the home.
Context-Aware Capabilities	Speech, pauses, topics, pace, stops, landmarks, time. Cannot see faces or distress: left to people.

Per-person settings

Agreed with the resident, recorded in their instance (TECH-04), changed whenever they ask.

Setting	Options	Default	Set by
Walks offered	alone; with listed partners	none until agreed	Resident with ST-05; ST-06 for walking alone
Guide on walks alone	silent; bridge to a person; conversation partner	bridge	Resident with ST-05 and ST-04
After a slip	wait only; wait, then place and thread	wait, then place and thread	Resident
Word help	off; after a wait	off	Resident
Starters	off; own life; shared ground with listed partners	own life	Resident; each starter consented per walk
Form of address	first name; surname; none	as the resident says	Resident
Pepper's welcome back	off; on	on	Resident
Research use of own words	off; on, anonymised	off	Resident and representative; ethics review
Walk length; rest offers	minutes	per resident	ST-06; the resident can shorten, not lengthen
Safety alerts	always on	on	Not adjustable
Sensitive topics	topic; trusted person; agreed approach	none	ST-05 with ST-04, family and ST-06

a3 Objective Stories

Abstract. Three short stories, each tying a Function to the value it serves, in a stakeholder's own words. Why the guide waits, then tells both (OS-01). Why painful questions go to a person (OS-02). Why the guide asks first (OS-03). Voices are provisional: written by the group, not yet heard from stakeholders.

Objective Story - Tell us both [OS-01]

Report Format: §2.2 Personas and Profiles

Fields

Field	Description
ID	OS-01
Name	Tell us both
Status	In Progress
Evidence basis	Provisional. Group reasoning; HFC-03, HFC-06.
Narrative	<i>Elena:</i> "At the counter I talked all day. Now a word goes, and the story goes with it. Give me a moment. If it doesn't come back, remind us both where I was, not just me."
Concerns	F2 Keep the thread; HV-02 Dignity, HV-06 Social connectedness

Objective Story - Fetch me [OS-02]

Report Format: §2.2 Personas and Profiles

Fields

Field	Description
ID	OS-02
Name	Fetch me
Status	In Progress
Evidence basis	Provisional. Group reasoning; HFC-08; IDP-03.
Narrative	<i>Myra:</i> "When Elena asks for Thomas, I want the guide to fetch me, not to answer her. Her son and I agreed what to say. One colleague told her bluntly, another said 'He's at work'; she trusted neither of us after that."
Concerns	F4 Defer sensitive topics; HV-08 Truthfulness, HV-05 Well-being

Objective Story - Not a dog [OS-03]

Report Format: §2.2 Personas and Profiles

Fields

Field	Description
ID	OS-03

Name	Not a dog
Status	In Progress
Evidence basis	Provisional. Group reasoning; VT-07.
Narrative	<i>Jan</i> : "I'm not a dog to be walked. I go out to see Elena, or the roses. The thing asks, and if I say no, that's that."
Concerns	F1 Ask first; HV-01 Autonomy, HV-02 Dignity

Graph edges

OS-01: narrates → F2, HV-02, HV-06; serves → OBJ-01, OBJ-07.

OS-02: narrates → F4, HV-08, HV-05; serves → OBJ-04.

OS-03: narrates → F1, HV-01, HV-02; serves → OBJ-06.

b1 Pair walk and walk alone

Abstract. Eight objectives, two use cases, six functions. UC01: a walk with a fellow resident, the main use case. UC02: a walk on your own. Six Premises verify that the Functions work; nine Claims, five of them adverse, test whether they help.

Objectives

Objective - Talk recovers [OBJ-01]

Field	Description
ID	OBJ-01
Name	Talk recovers
Status	In Progress
Evidence basis	Goal on SA-02; PS-01. HFC-03, HFC-05.
Statement	After a slip, the conversation goes on, without the slip being exposed.
Type	Human-factors
Derived from	Goal on SA-02; HFC-03 Conversation repair
Priority	Must.

Objective - Walk with company [OBJ-02]

Field	Description
ID	OBJ-02
Name	Walk with company
Status	In Progress
Evidence basis	Goals on SA-01, SA-02. HFC-01. Forbes et al. (2015).
Statement	Residents walk more often with a fellow resident, or with the guide where agreed.
Type	Operational
Derived from	Goals on SA-01, SA-02; HV-06
Priority	Must.

Objective - Safe while talking [OBJ-03]

Field	Description
ID	OBJ-03
Name	Safe while talking
Status	In Progress
Evidence basis	VT-01; HFC-02; PS-04. Lundin-Olsson et al. (1997).
Statement	Talk on the walk does not raise falls or let tiredness pass unnoticed.
Type	Value-derived

Derived from	HV-04 Safety; bounded by VT-01
Priority	Must.

Objective - No deception [OBJ-04]

Field	Description
ID	OBJ-04
Name	No deception
Status	In Progress
Evidence basis	PS-02; VT-03; HFC-08.
Statement	The System never lies, denies or plays along, and corrects its own errors at once. Painful truths come from a trusted person.
Type	Value-derived
Derived from	HV-08 Truthfulness
Priority	Must.

Objective - No secret revealed [OBJ-05]

Field	Description
ID	OBJ-05
Name	No secret revealed
Status	In Progress
Evidence basis	VT-04, VT-05; PS-05.
Statement	Nothing about a resident reaches another person without that resident's consent. Nothing leaves the care home.
Type	Value-derived
Derived from	HV-03 Privacy
Priority	Must.

Objective - The resident's own walk [OBJ-06]

Field	Description
ID	OBJ-06
Name	The resident's own walk
Status	In Progress
Evidence basis	VT-07; HV-01; PS-03. Ryan & Deci (2000).
Statement	Walks happen for the resident's own reasons, with the resident's own settings; never as a routine run for staff.
Type	Value-derived
Derived from	HV-01 Autonomy; bounded by VT-07
Priority	Must.

Objective - Adult voice [OBJ-07]

Field	Description
ID	OBJ-07
Name	Adult voice
Status	In Progress
Evidence basis	HFC-06. Williams et al. (2009); Kitwood (1997).
Statement	No elderspeak, no quizzing, no correcting, no praise for the obvious.
Type	Value-derived
Derived from	HV-02 Dignity
Priority	Should.

Objective - Less lonely [OBJ-08]

Field	Description
ID	OBJ-08
Name	Less lonely
Status	In Progress
Evidence basis	HFC-07; PS-06. Victor (2012); Holwerda et al. (2014).
Statement	Over weeks, residents feel less lonely.
Type	Human-factors
Derived from	HV-06 Social connectedness; HFC-07
Priority	Should. Exploratory within this project's time frame.

Graph edges

OBJ-01: derived-from → Goal (SA-02), HFC-03; served-by → F2; referenced-by → UC01-F2-CL1, UC01-F2-CL3; narrated-by → OS-01.

OBJ-02: derived-from → Goals (SA-01, SA-02), HV-06; served-by → F1, F2; referenced-by → UC01-F1-CL5.

OBJ-03: derived-from → HV-04, HFC-02; served-by → F3, F5; referenced-by → UC01-F3-CL4.

OBJ-04: derived-from → HV-08, HFC-08; served-by → F2, F4; referenced-by → UC02-F4-CL6; narrated-by → OS-02.

OBJ-05: derived-from → HV-03; served-by → F1, F6; referenced-by → UC01-F2-CL8.

OBJ-06: derived-from → HV-01; served-by → F1; referenced-by → UC01-F1-CL7; narrated-by → OS-03.

OBJ-07: derived-from → HV-02; served-by → F2; referenced-by → UC01-F2-CL2; narrated-by → OS-01.

OBJ-08: derived-from → HV-06, HFC-07; served-by → F1, F2; referenced-by → UC01-F1-CL9.

Use Case - A walk with a fellow resident [UC01]

Report Format: §2.3 Objectives, Use Cases, Functions, Premises and Claims

Fields

Field	Description
-------	-------------

ID	UC01
Name	A walk with a fellow resident
Status	In Progress
Evidence basis	Formalizes SA-02, PS-01, PS-03 to PS-05 and DS-01. No primary data.
Objective(s) served	OBJ-01 to OBJ-08
TDP(s) and IDP(s) used	TDP-01; IDP-01, IDP-02, IDP-03
Actors	- ST-01 Resident and ST-02 Fellow resident (Direct) - walk, talk, consent, end the walk - ST-04 Care worker (Direct) - clips on the guides; answers alerts and listed topics - ST-05 Activity coordinator (Indirect) - pairs, settings, relationship map - System - Pepper (TECH-01), a pocket guide each (TECH-02), beacons (TECH-03), server and exchange (TECH-04, TECH-05), walker model (TECH-06)
Trigger	A pair's agreed time; or one resident asks for a walk with a listed partner.
Pre-condition	- Both assessed fit for the garden loop with a partner (ST-06) - The pair in the map, agreed by both; no map change awaiting review - Guides charged; server up; staff reachable - Consent arrangements in place (O-5)
Post-condition	Both back inside, or a care worker has taken over after F5. Record: six items. Thread, transcript and exchanged items deleted.
Action Sequence	1. Each resident asked alone, by Pepper or by the guide in their room: a walk with this partner; each planned starter (F1, IDP-02). 1-alt. Either says no: no pair walk. The other hears "Not today" and is offered a walk alone, or another time. No reason given or asked. 2. Instances exchange consented items the map allows (F1). Guides report ready, or degraded (F5). 3. Pepper introduces both by name and photo (F2). 4. Walk. Guides silent while the pair talks; walker model updated (F2). 5. Slip: wait; then place, thread or a consented shared topic, as a statement to both (F2, IDP-01). Starters only at a stop (F3). 6. Belief from another time: the feeling, then the present (F2, IDP-01). Question on a listed topic: truthful deferral; trusted person alerted (F4, IDP-03). 7. Stumble while talking: offer to sit. Second within 10 min: alert (F3, F5). 8. One walker presses "going in": the other is offered to go on alone, if assessed fit, or to go in too (F1). 9. Set length reached: head-back cue, even mid-talk (F3). 10. At the door, Pepper's welcome back: one statement from the walk (F2). 11. Record kept; thread, transcript and exchanged items deleted (F6). Exception. Guide, server or map unavailable: the walk goes on as today, without personal starters. Degraded state shown at step 2.
Function(s) realized	F1 to F6

Use Case - A walk on your own [UC02]

Report Format: §2.3 Objectives, Use Cases, Functions, Premises and Claims

Fields

Field	Description
ID	UC02
Name	A walk on your own
Status	In Progress
Evidence basis	Formalizes SA-01, PS-02, PS-06 and DS-02. No primary data.
Objective(s) served	OBJ-02 to OBJ-08
TDP(s) and IDP(s) used	TDP-01; IDP-01, IDP-02, IDP-03
Actors	- ST-01 Resident (Direct) - walks; talks with the guide where agreed - ST-04 Care worker (Direct) - trusted person; answers alerts - System - pocket guide (TECH-02), beacons (TECH-03), server and map (TECH-04, TECH-05), walker model (TECH-06)
Trigger	The resident goes out.
Pre-condition	Assessed fit to walk alone (ST-06). Guide setting for walks alone agreed with resident and staff.
Post-condition	Resident back inside, or a person has come after F4 or F5. Record: six items.
Action Sequence	- At the door, the guide asks consent for today's own-life starters (F1). - Guide follows the setting: silent; bridge; or conversation partner, with short replies on the resident's own topics and place cues with long silences (F2). - Bridge: at a bench, "Shall I ask [partner] about a walk tomorrow?" (F1). - Question on a listed topic: truthful deferral; trusted person alerted (F4, IDP-03). If the resident declines, the trusted person is told after the walk. - Tiredness, a stumble, the gate: rest offer; alert (F3, F5). - Record kept (F6).
Function(s) realized	F1 to F6

Functions

Function	Behaviour	Serves	Bounded by	Enabled by
F1 Ask first	Before a walk, each walker alone: walk with this person; each starter; listening (study only). Two options; "no" ends the offer for the walk; no reason asked. On walks alone, offers a person (bridge).	OBJ-02, OBJ-05, OBJ-06	VT-04, VT-07	TECH-01, TECH-02, TECH-05
F2 Keep the thread	Walker model: common ground, thread, said already, current belief, state. After a slip: wait, then place, thread or a consented shared topic, as a statement to both (IDP-01). Belief from another time: the feeling, then the present. Silent while people talk. Contradicted: a plain correction ("I had that wrong"); the error flagged for ST-05.	OBJ-01, OBJ-02, OBJ-04, OBJ-07	VT-02, VT-05	TECH-02, TECH-03, TECH-06
F3 Watch pace and rest	Starters only at a stop. No questions while walking. Stumble: offer to sit; second: F5. Pace drop or set length: rest offer; head-back cue, even mid-talk.	OBJ-03	VT-01	TECH-02, TECH-03
F4 Defer sensitive topics	Question on a listed topic: "That's one for [trusted person]. Shall I ask them to come?" Trusted person alerted. Never states the fact; never denies it.	OBJ-04	VT-03	TECH-04, TECH-05, TECH-06

F5 Call for help	Alert on pull-cord, gate, fall-like impact, second stumble, no response. Tells the resident someone is coming. Reports a degraded state at the start.	OBJ-03	VT-06	TECH-02 to TECH-04
F6 Keep a short record	Per walk: type, minutes, ended early, stumbles, person called and reason, exchanged item IDs. Deletes thread, transcript and exchanged items at the end. Nothing leaves the home.	OBJ-05	VT-04, VT-05	TECH-04

Why sentences are prepared, never generated. A generated sentence can state something untrue about a person's life, and cannot be checked before it is said (HV-08). The coordinator and family check the prepared lines (O-4).

Premises

Verification only: does the Function work as specified.

Premise ID	Function	Evaluation Method(s) / Measure(s)	Verification Criteria	Step(s)
UC01-F1-PR1	F1	EM-01; M-01, M-10. No human subjects.	No pair walk, exchange or starter without a yes from its owner, asked alone, this walk. "No" ends the offer for the walk. Blocked or off-limits items never exchanged: 0/50.	UC01 1-2; UC02 1, 3
UC01-F2-PR2	F2	EM-01; M-01. No human subjects.	No System speech within [8 s - O-2] of a mid-sentence stop; echo within [3 s - O-2] after. Script check: statements only; no untruth, denial or playing along; none of "What were you saying", "You told", "Do you remember". Contradicted: plain correction ≤ 3 s, error flagged.	UC01 3-6, 10; UC02 2
UC01-F3-PR3	F3	EM-01; M-01. No human subjects.	No starter while moving: 0/20. Stumble response ≤ 3 s. Head-back cue ≤ 30 s after the set length.	UC01 5, 7, 9; UC02 5
UC02-F4-PR4	F4	EM-01; M-01, M-10. No human subjects.	Listed sensitive questions detected in 19/20. Response is a truthful deferral; trusted person alerted ≤ 10 s. Listed fact never stated or denied: 0/50.	UC01 6; UC02 4
UC01-F5-PR5	F5	EM-01; M-01. No human subjects.	Gate, pull-cord, second stumble, drop with no response: alert ≤ 10 s, 20/20. Unacknowledged alerts escalate. Degraded state reported at the start.	UC01 2, 7; UC02 5
UC01-F6-PR6	F6	EM-01; M-01 (content, network and file checks). No human subjects.	Record holds exactly six items. Thread, transcript and exchanged items deleted at the end. No packet leaves the home network; no audio file on any device.	UC01 11; UC02 6

Claims

Validation: does the Function have the intended effect.

Claim ID	Function	Serves Objective	Effect	Grounding	Evaluation Method(s) / Measure(s)	Validation Criteria	Step(s)
UC01-F2-CL1	F2	OBJ-01	Positive. Wait-then-redirect raises the share of slips recovered on pair walks, compared with walks as today. Concerns ST-01, ST-02, SA-02, PS-01.	HFC-03, HFC-04, HFC-05; Schegloff et al. (1977); Orange et al. (1996).	EM-03; M-04, M-03	NAP $\geq .70$ at all three phase changes in 4 of 6 pairs; face signs (M-04) and negative emotion (M-03) no higher in B.	5

UC01-F2-CL2	F2	OBJ-07	Adverse. Help after a slip is felt as being corrected or exposed. Concerns ST-01, HV-02.	HFC-06, HFC-07; Goffman (1967).	EM-03, EM-04; M-04 (face signs), M-09 (T)	Confirmed if face signs per slip rise in B, or residents describe the help as talking down (code T). If confirmed: "wait only" for that person.	5
UC01-F2-CL3	F2	OBJ-01	Adverse. The guide cuts in before the speaker can repair. Concerns ST-01, SA-02.	HFC-03; Schegloff et al. (1977).	EM-03; M-04 (self-repair share)	Confirmed if the self-repair share falls in B. If confirmed: a longer wait, per person.	5
UC01-F3-CL4	F3	OBJ-03	Adverse. Talk kept going on the walk raises stumbles, or lets walks run past tiredness. Concerns ST-01, HV-04, PS-04.	HFC-02; Lundin-Olsson et al. (1997).	EM-03; M-06	Confirmed if stumbles per walk-hour are higher in B than in A in 2 or more pairs, or overruns rise. If confirmed: starters only when seated.	5, 7, 9
UC01-F1-CL5	F1	OBJ-02	Positive. Asking first, with introductions, leads to more walks with company and more pleasure on them. Concerns ST-01, ST-02, SA-02.	HFC-01; Ryan & Deci (2000).	EM-03; M-02, M-03	Walks with company per week, and the positive emotion score, higher in B in 4 of 6 pairs.	1, 3
UC02-F4-CL6	F4	OBJ-04	Positive. Painful questions are answered by the trusted person, and trust is kept: no complaint of being misled or fobbed off. Concerns ST-01, ST-04, SA-01, PS-02.	HFC-08; Mental Health Foundation (2016).	EM-03, EM-04; M-10, M-09 (U)	All listed questions deferred; trusted person arrives within [threshold - O-2]; System errors below [threshold - O-2] per 100 statements, all corrected; no U codes.	UC01 6; UC02 4
UC01-F1-CL7	F1	OBJ-06	Adverse. The walk is felt as being "taken out", for staff's sake. Concerns ST-01, ST-03, VT-07, PS-03.	VT-07; Kitwood (1997).	EM-03, EM-04; M-09 (D), M-07 ("Again?")	Confirmed if residents or relatives use code D, or the "Again?" share falls in B. If confirmed: offers only at the resident's request.	1
UC01-F2-CL8	F2	OBJ-05	Adverse. A guide that listens is felt as being listened in on. Concerns ST-01, ST-02, HV-03, VT-05.	VT-05; Niemeijer et al. (2015).	EM-04; M-09 (L, W)	Confirmed if residents or partners use codes L or W. If confirmed: thread-keeping off by default, on at the resident's request.	4
UC01-F1-CL9	F1, F2	OBJ-08	Positive, exploratory. Loneliness is lower at the end of B phases than of A phases. Concerns ST-01, HV-06, PS-06.	HFC-07; Holwerda et al. (2014).	EM-03; M-08	Lower in B in 4 of 6 pairs. Phases are short: a null result is likely and reported as such.	-

Graph edges

UC01: realizes → DS-01; serves → OBJ-01 to OBJ-08; uses → TDP-01, IDP-01 to IDP-03; asserts → UC01-F1-PR1, UC01-F2-PR2, UC01-F3-PR3, UC01-F5-PR5, UC01-F6-PR6, UC01-F2-CL1, UC01-F2-CL2, UC01-F2-CL3, UC01-F3-CL4, UC01-F1-CL5, UC01-F1-CL7, UC01-F2-CL8, UC01-F1-CL9.

UC02: realizes → DS-02; serves → OBJ-02 to OBJ-08; uses → TDP-01, IDP-01 to IDP-03; asserts → UC02-F4-PR4, UC02-F4-CL6.

Open items

ID	Item	Blocks	Owner
O-1	The group observation behind HFC-01 needs a direct source.	HFC-01	Group
O-2	Thresholds: wait length, echo latency, stumble response, alert latency, trusted person's arrival, System error rate. Set from an EM-01 pilot and with care workers.	PR2 to PR5; CL6	Group
O-3	Offline speech recognition of older speakers, in their own language and dialect, in wind. Test in EM-01.	TECH-06; F2, F4	Group

O-4	Relationship map and prepared lines: form, owner, review after events; how staff learn of conflicts between residents.	F1, F2, F4; VT-04	ST-05 with group
O-5	Consent and data: GDPR art. 6 and 9; representative; process consent per walk; check of the local medical-research law.	All pre-conditions; EM-02, EM-03	ST-07
O-6	Garden policy with ST-07 and ST-06: who may walk alone or in pairs; who answers alerts.	UC01, UC02 pre-conditions; VT-06	ST-07, ST-06
O-7	Research use of own-word lists: anonymisation; re-identification risk in a small home with a local dialect; ethics review.	EM-02 language notes; VT-05	Group
O-8	Pepper at the door: a private spot for consent questions, out of the walking path. Check on site.	TECH-01; RP-01	Group

References

Cited on this page: Forbes et al. (2015); Goffman (1967); Holwerda et al. (2014); Kitwood (1997); Lundin-Olsson et al. (1997); Mental Health Foundation (2016); Niemeijer et al. (2015); Orange et al. (1996); Ryan & Deci (2000); Schegloff et al. (1977); Victor (2012); Williams et al. (2009). Full entries on pages a1.1, b1.1, b3.1 and b4.1.

c1 Thread in the pocket, person in reach

Abstract. One pattern: who repairs, who tells, who keeps safe. The speaker gets the first chance to repair, the companion the second, the System the third. Painful truths go to a trusted person. Safety stays with staff, whom the System calls. Consent and settings stay with the resident.

Team Design Pattern - Thread in the pocket, person in reach [TDP-01]

Report Format: §2.4 Team Design Patterns (Task Level)

Fields

Field	Description
ID	TDP-01
Name	Thread in the pocket, person in reach
Status	In Progress
Evidence basis	Literature and group reasoning only. Not yet evaluated. - Derived from SA-01, SA-02, PS-01 to PS-06, UC01, UC02 - Order of repair: HFC-03 (Schegloff et al., 1977) - Partners miss each other's slips: HFC-05 (Bora et al., 2015) - Painful truths from a trusted person: HFC-08
Problem	People with dementia lose the thread of a conversation. A partner with dementia often misses it; staff cannot walk every walk. Painful questions go to whoever is near. Who does what?
Solution Structure	1. Speaker: first chance to repair. 2. Companion (fellow resident): talk, repair, comfort. 3. System: keeps the thread, waits, redirects gently, watches pace, asks consent, defers. 4. Trusted staff: painful truths; physical help. 5. Coordinator: pairs, settings, relationship map.
Solution Description	The System keeps the thread outside the walkers' heads, and offers it back only after the speaker has had a moment. Anything painful or physical goes to a person: the System calls, it does not tell.

Per-Actor allocation	Resident (ST-01, ST-02): 1. Consent, or not; set own preferences. 2. Talk; repair own slips first. 3. Pause the guide; end the walk. Relative (ST-03): 1. Tell the life story; list welcome and off-limits topics. 2. Agree with staff what is said about painful topics. 3. Consent on the resident's behalf, where needed. Care worker (ST-04): 1. Clip on the guides; check they report ready. 2. Answer alerts. 3. Answer listed topics as agreed with family. Activity coordinator (ST-05): 1. Agree settings with each resident. 2. Pair residents, with both consenting. 3. Keep the relationship map current with ST-04 and family. System: 1. F1: ask first. 2. F2: keep the thread; redirect gently. 3. F3: watch pace and rest. 4. F4: defer listed topics. 5. F5: call for help. 6. F6: six-item record.
Consequences: benefits	Conversations recover without exposing the slip. Walks with company need less staff time. Painful news comes from the same trusted person each time. Expected effects, still to be tested: UC01-F2-CL1, UC01-F1-CL5, UC02-F4-CL6.
Consequences: risks	• Companions rely on the guide and stop helping • The guide cuts in before self-repair (UC01-F2-CL3) • Help after a slip feels like correction (UC01-F2-CL2) • An out-of-date map lets a secret through, or misses a sensitive topic
Related Patterns	IDP-01 Wait, then redirect gently; IDP-02 Ask first; IDP-03 Truthful deferral.
Connection to Claims	Rationale for UC01-F2-CL1, UC01-F1-CL5, UC02-F4-CL6. Tested against UC01-F2-CL2, UC01-F2-CL3.
Validation path	EM-01 (the hand-offs work), EM-03 (slips recovered, per case), EM-04 (residents, relatives and staff accept the split).

Open items to check or research

Item	What is needed	Reason
Relationship map (O-4)	Form, owner, review after every event.	An out-of-date map is the main route to a disclosure.
Garden policy (O-6)	Agree with ST-07 and ST-06 who may walk alone or in pairs.	Without it, the allocation has no permission.
Companion briefing	One page for walking partners and families: what the guide does, and what it never does.	Companions must not assume the guide watches over the resident.

Graph edges

allocates → SA-01, SA-02 tasks to ST-01 to ST-05; F1 to F6 to System; supports → UC01-F2-CL1, UC01-F1-CL5, UC02-F4-CL6; tested-against → UC01-F2-CL2, UC01-F2-CL3.

References

- Bora et al. (2015); Schegloff et al. (1977). See a1.1 References.

d1 How the guide speaks

Abstract. Three patterns for how the guide speaks. After a slip, wait, then redirect gently (IDP-01). Ask the owner, alone, before joining or using a starter (IDP-02). Never deceive, correct errors at once, and pass painful truths to a trusted person (IDP-03). Wording and timing are hypotheses until EM-01 Build A.

Interaction Design Pattern - Wait, then redirect gently [IDP-01]

Report Format: §2.5 Interaction Design Patterns (Interaction Level)

Fields

Field	Description
ID	IDP-01
Name	Wait, then redirect gently
Status	In Progress
Evidence basis	Literature and group reasoning only. Derived from SA-02, PS-01, DS-01. Grounded in HFC-01, HFC-03, HFC-04, HFC-06. Not yet evaluated.
Problem	After a slip, help too soon takes the turn; help too late lets the talk die. A correction or a quiz exposes the slip.
Principle	The speaker gets the first chance. Then offer both walkers a way back, as a statement.
Solution	1. Wait: about 8 s after a mid-sentence stop (O-2); keep walking. 2. Place: something here and now. "The pond's just ahead." 3. Thread, to both: "Elena was telling you about pension day." 4. Shared topic, consented this walk: "Elena had the post office, and Jan the bakery, in the same town." 5. Belief from another time: the feeling, then the present. "You're thinking of your mother. The sun is on the bench." No correction; no confirmation of what is untrue. 6. Word help only if the person's setting allows it, after the wait, as a guess to both: "The sorting office?" 7. Never: "What were you saying?", "You told us that already", "Do you remember...?" 8. Silent again as soon as the talk resumes.
Research Link	HFC-03: speakers prefer self-repair (Schegloff et al., 1977); repair fails more often in dementia (Orange et al., 1996). HFC-04: common ground. HFC-06: face. HFC-01: the place as a topic that asks nothing of memory.
Connection to Claims	Shapes F2 (UC01 steps 5, 6; UC02 step 2). Rationale for UC01-F2-CL1. Tested against UC01-F2-CL2, UC01-F2-CL3.

Graph edges

shapes → F2; supports → UC01-F2-CL1; tested-against → UC01-F2-CL2, UC01-F2-CL3; informed-by → EM-01 (Build A).

Interaction Design Pattern - Ask first [IDP-02]

Report Format: §2.5 Interaction Design Patterns (Interaction Level)

Fields

Field	Description
ID	IDP-02
Name	Ask first
Status	In Progress
Evidence basis	Group reasoning. Derived from SA-02, PS-03, PS-05, VT-04, VT-07, DS-01. Grounded in HV-01, HV-03; Ryan & Deci (2000); process consent (Dewing, 2007). Not yet evaluated.
Problem	A personal starter reveals something about its owner. A partner who asks is hard to refuse. Asked in front of the other walker, "no" is hard to say.
Principle	Ask the owner, alone, before. Two options. "No" is final.
Solution	1. Ask each walker alone, before the walk: "Elena would like a walk with you. Join her, or not today?" 2. Ask the owner of each starter, alone: "May I mention the post office counter to Jan?" 3. Consent lasts one walk. 4. After a no: no reason asked. The other walker hears only "Not today." 5. Never ask in front of the other walker. 6. No reminders of missed walks; no tallies.
Research Link	Self-determination theory: choice supports motivation (Ryan & Deci, 2000). Process consent: consent in dementia is renewed at each contact (Dewing, 2007).
Connection to Claims	Shapes F1. Rationale for UC01-F1-CL5. Tested against UC01-F1-CL7. Verified by UC01-F1-PR1.

Graph edges

shapes → F1; supports → UC01-F1-CL5; tested-against → UC01-F1-CL7; verified-by → UC01-F1-PR1.

Interaction Design Pattern - Truthful deferral [IDP-03]

Report Format: §2.5 Interaction Design Patterns (Interaction Level)

Fields

Field	Description
ID	IDP-03
Name	Truthful deferral
Status	In Progress
Evidence basis	Literature and group reasoning only. Derived from SA-01, PS-02, DS-02. Grounded in HV-08, HFC-08. Not yet evaluated.
Problem	Painful questions come anywhere. A kind lie costs trust. A blunt truth from a stranger, or a machine, causes grief.
Principle	Never deceive; correct errors at once. Pass painful truths to the trusted person.

Solution	1. Listed topic (relationship map): "That's one for Myra. Shall I ask her to come?" 2. Alert the trusted person: resident, topic, place. 3. No answer to the question itself; no denial. 4. If the resident declines: the trusted person is told after the walk. 5. The guide does not raise the topic again. 6. Unlisted but possibly painful (a death, an illness, going home): treated as listed; the trusted person is the key worker. 7. The trusted person decides what is said, as agreed with family and ST-06. 8. Contradicted, or found wrong: "I had that wrong." No argument; the error flagged for ST-05 to fix at its source.
Research Link	HFC-08: lies found out cost trust (Mental Health Foundation, 2016; James et al., 2006).
Connection to Claims	Shapes F4. Rationale for UC02-F4-CL6. Verified by UC02-F4-PR4.

Graph edges

shapes → F4; supports → UC02-F4-CL6; verified-by → UC02-F4-PR4.

Open items to check or research

Item	What is needed	Reason
Wait and wording (O-2)	EM-01 Build A: wait length, thread wording, word-help form.	Every pattern here is a hypothesis about how a voice should sound.
Relationship map (O-4)	Form and review routine for listed topics and trusted persons.	IDP-02 and IDP-03 are only as safe as the map behind them.